

Biot Savart Law

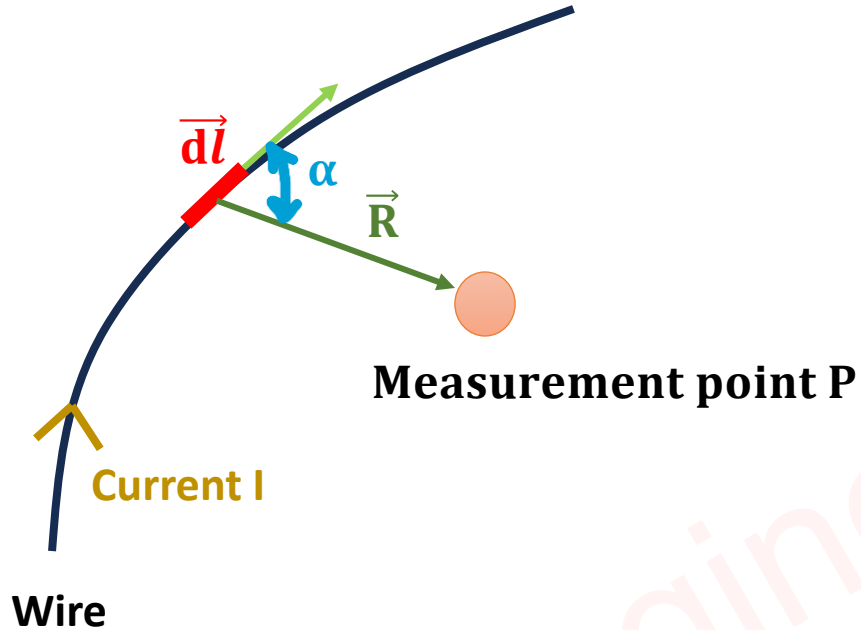
Outlines

- Basics of Biot Savart Law
- Direction of Magnetic Field Intensity
- Case Study of Magnetic Field Intensity

Engineering Funda

Biot Savart Law

Basics of Biot Savart Law



- Magnetic Field Intensity dH due to small element is given by :

$$\Rightarrow dH \propto \frac{Idl \sin \alpha}{R^2}$$

$$\Rightarrow dH = \frac{1}{4\pi} \frac{Idl \sin \alpha}{R^2}$$

- Biot Savart Law states that the differential Magnetic field intensity dH produced at point P by the differential current element Idl is directly proportional to product of Idl and sine of angle between $d\vec{l}$ and position vector $\vec{R}$ and it is inversely proportional to square of distance between point P and element $d\vec{l}$.

Direction of Magnetic Field Intensity

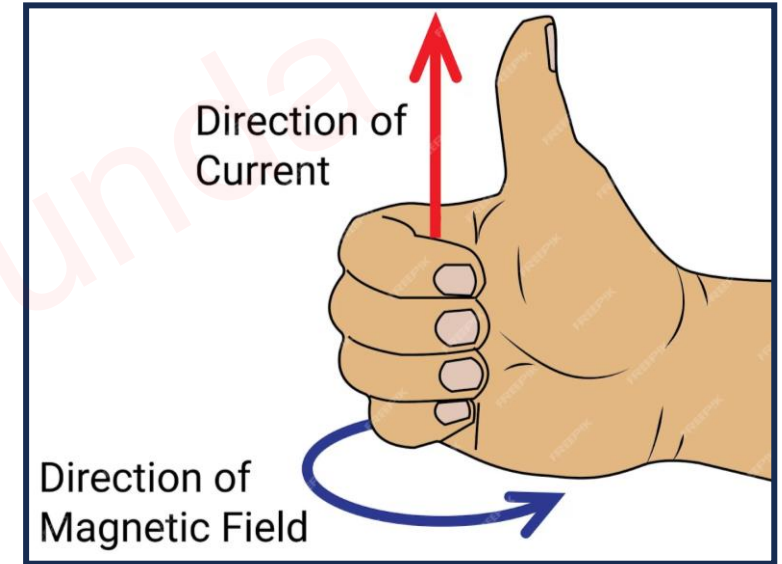
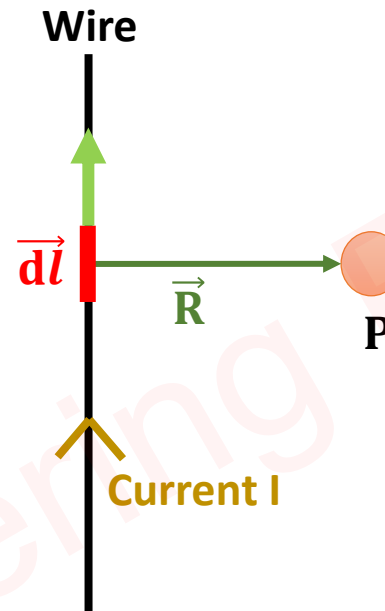
- Magnetic Field Intensity due to small element is given by :

$$\Rightarrow dH \propto \frac{Idl \sin \alpha}{R^2}$$

$$\Rightarrow dH = \frac{1}{4\pi} \frac{Idl \sin \alpha}{R^2}$$

$$\Rightarrow d\vec{H} = \frac{1}{4\pi} \frac{Id\vec{l} \times \hat{R}}{R^2}$$

$$\Rightarrow d\vec{H} = \frac{1}{4\pi} \frac{Id\vec{l} \times \vec{R}}{R^3}$$



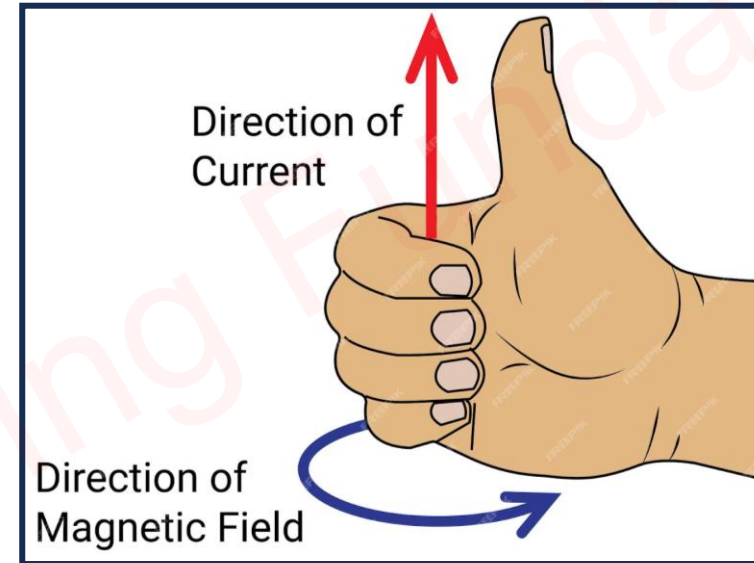
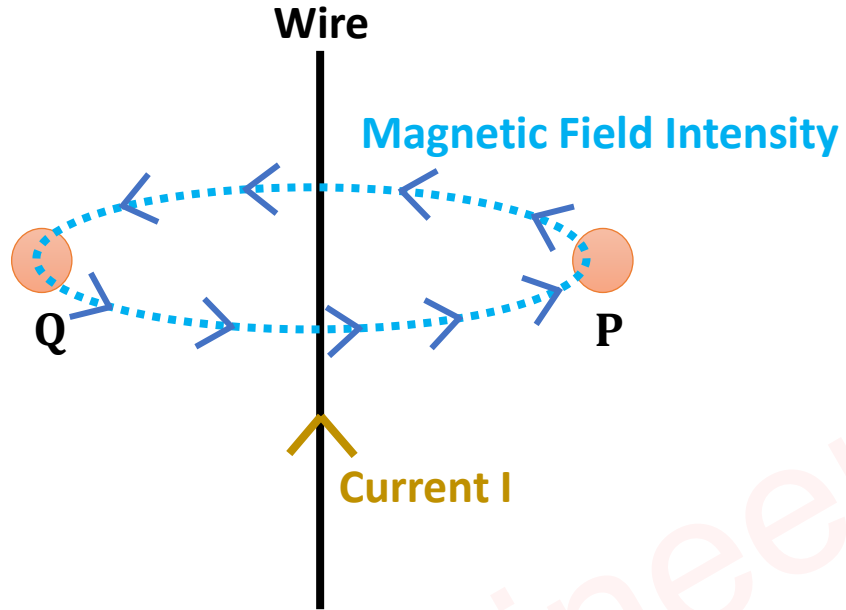
- Direction of Magnetic Field Intensity is as per right hand thumb rule between $Id\vec{l}$ and $\vec{R}$.
- Total Magnetic Field Intensity is given by:

$$\Rightarrow \vec{H} = \int d\vec{H} = \int \frac{1}{4\pi} \frac{Id\vec{l} \times \vec{R}}{R^3}$$

- Unit of Magnetic Field Intensity is A/m.

Case Study of Magnetic Field Intensity

- Find Direction of Magnetic Field Intensity at Point P and Point Q.



- At Point P, the Direction of the Magnetic field Intensity is inward to the page.
- At Point Q, the Direction of the Magnetic field Intensity is outward from the page.

Ampere's Circuit Law

Outlines

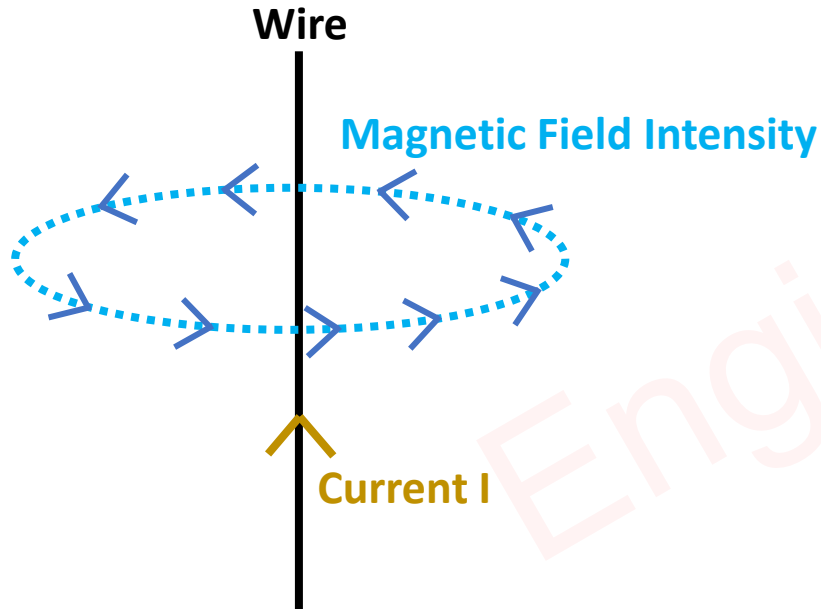
- Basics of Ampere's Circuit Law
- Case Study of Ampere's Circuit Law
- Magnetic Field Intensity due to straight long wire using Ampere's Circuit Law

Ampere's Circuit Law

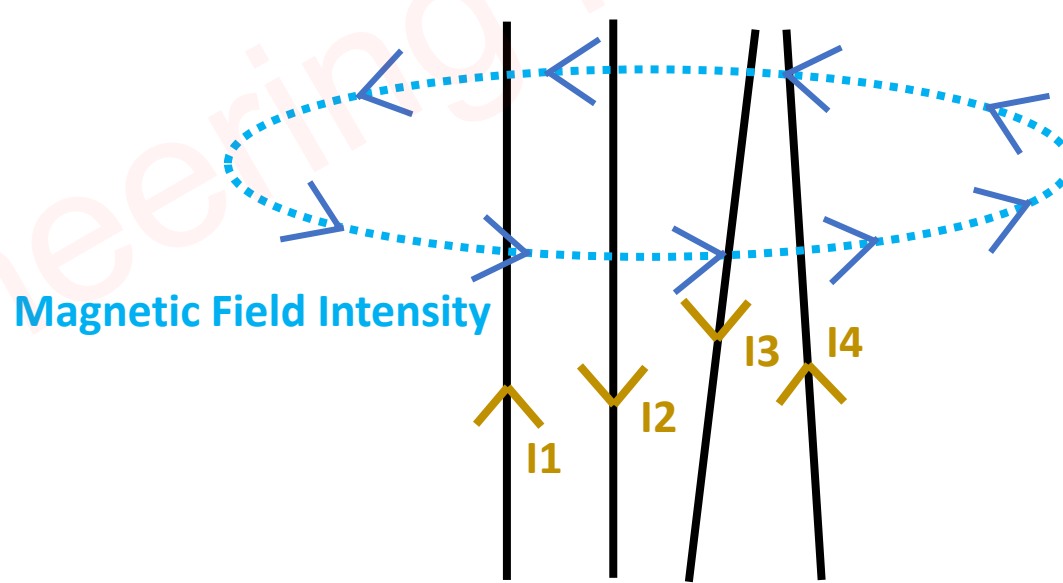
Basics of Ampere's Circuit Law

- Ampere's Law states that "Line Integrals of Magnetic Field Intensity over Closed Loop is equal to Total Current enclosed by that loop."

$$\Rightarrow \oint \vec{H} \cdot d\vec{l} = I$$



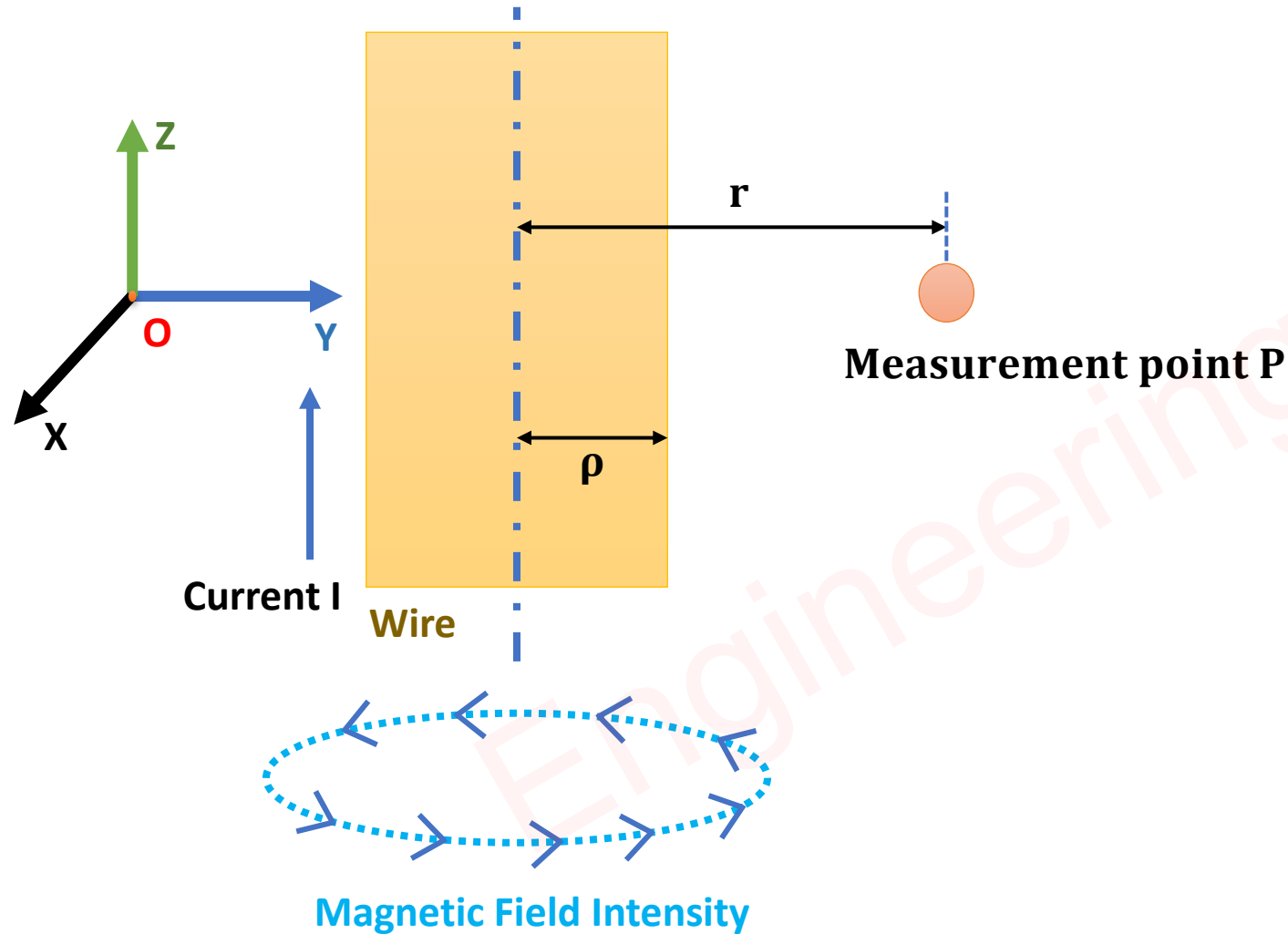
$$\Rightarrow \oint \vec{H} \cdot d\vec{l} = I$$



$$\Rightarrow \oint \vec{H} \cdot d\vec{l} = (I_1 - I_2 - I_3 + I_4)$$

Magnetic Field Intensity for Straight Long Wire

- Case – I: Magnetic Field Intensity Outside the wire: ($r > \rho$)



$$\Rightarrow \oint \vec{H} \cdot d\vec{l} = I$$

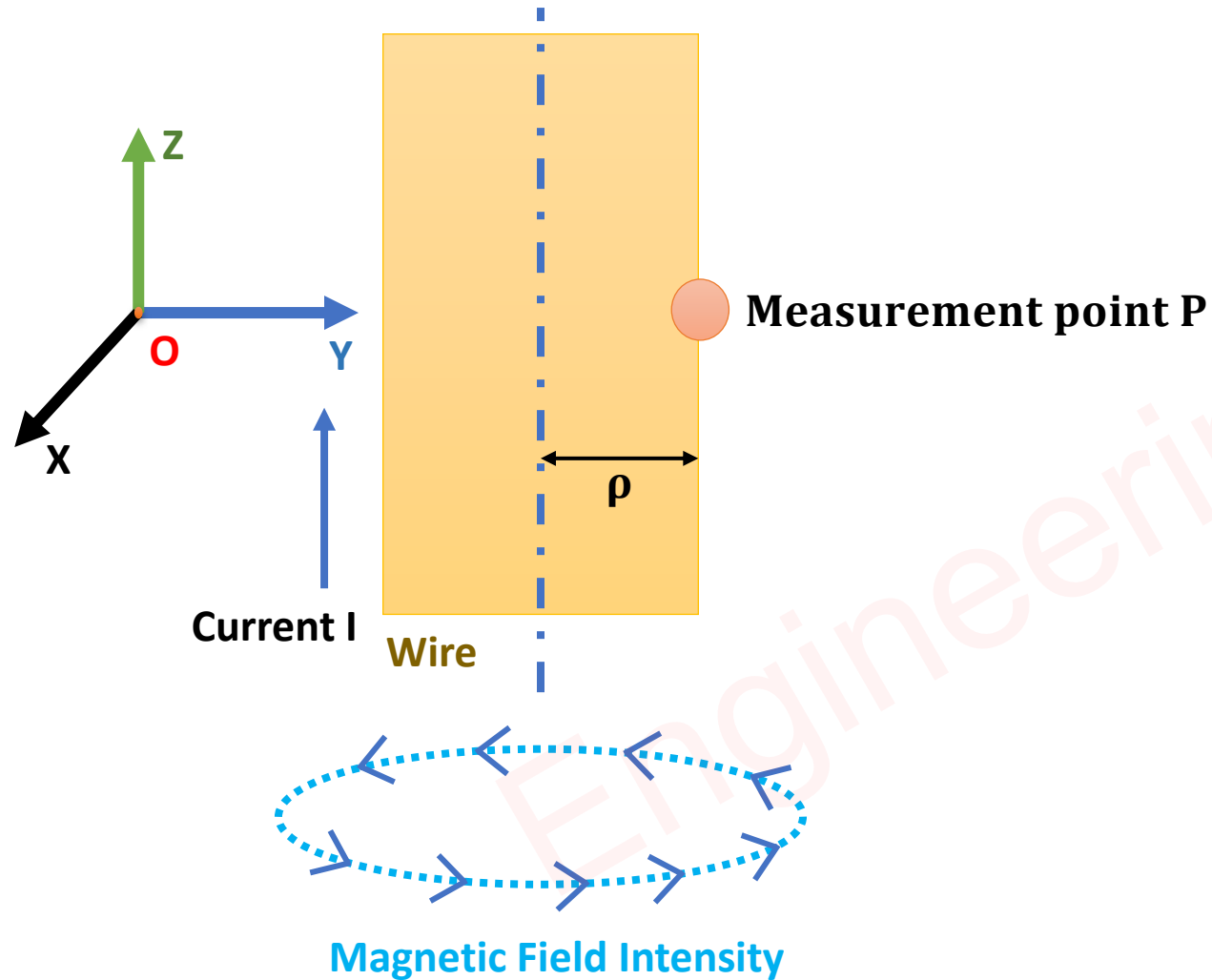
$$\Rightarrow H(2\pi r) = I$$

$$\Rightarrow H = \frac{I}{2\pi r}$$

$$\Rightarrow \vec{H} = \frac{I}{2\pi r} \hat{i}_\phi$$

Magnetic Field Intensity for Straight Long Wire

- Case – II: Magnetic Field Intensity on surface of the wire: ($r = \rho$)



$$\Rightarrow \oint \vec{H} \cdot d\vec{l} = I$$

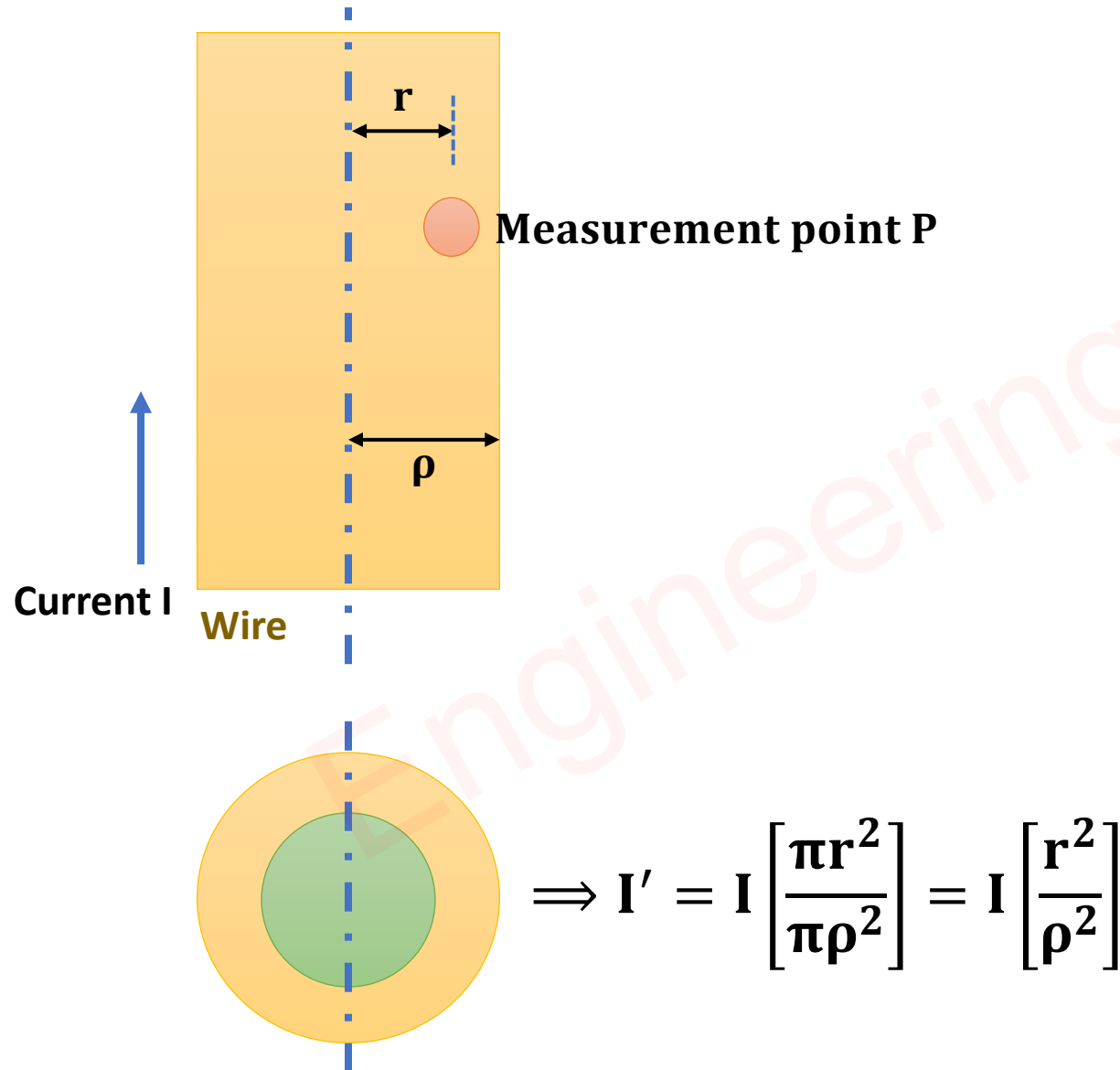
$$\Rightarrow H(2\pi\rho) = I$$

$$\Rightarrow H = \frac{I}{2\pi\rho}$$

$$\Rightarrow \vec{H} = \frac{I}{2\pi\rho} \hat{i}_\phi$$

Magnetic Field Intensity for Straight Long Wire

- Case – III: Magnetic Field Intensity inside the wire: ($r < \rho$)



$$\Rightarrow \oint \vec{H} \cdot d\vec{l} = I'$$

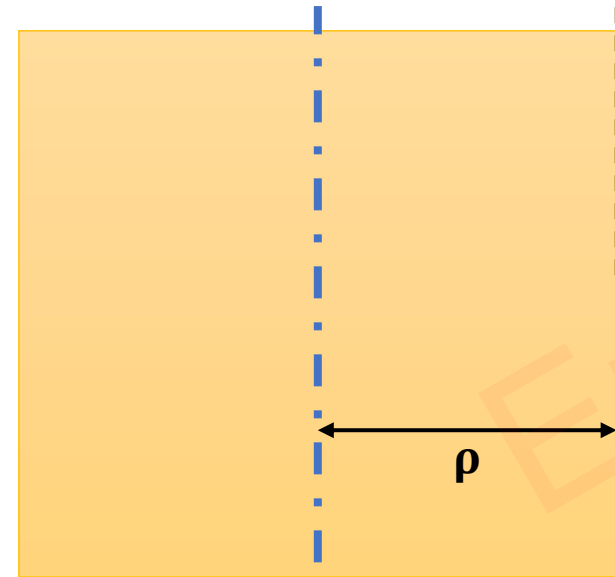
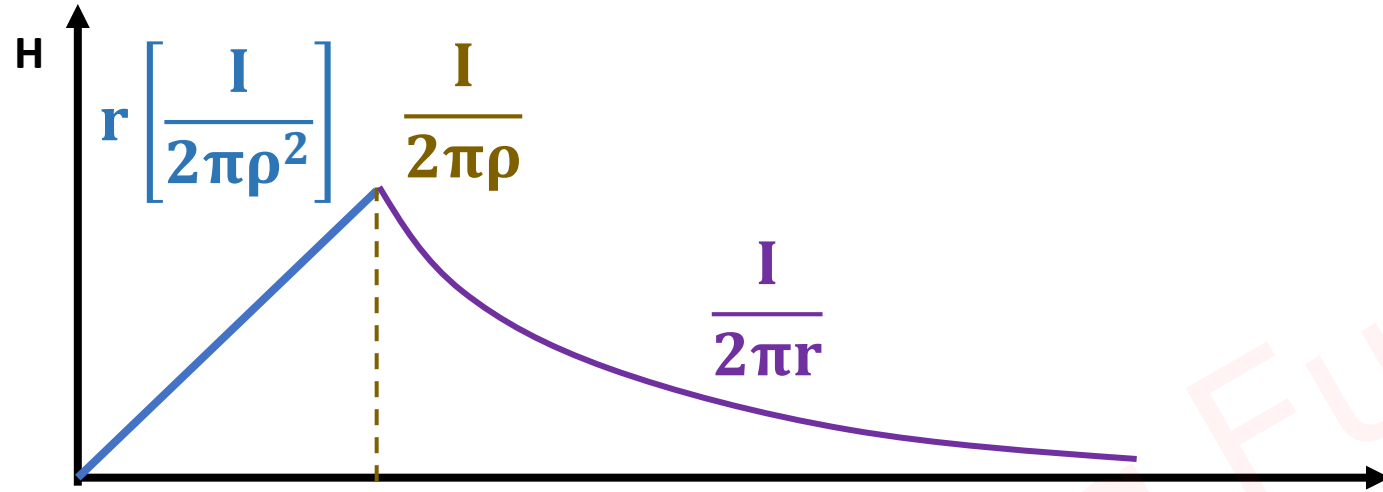
$$\Rightarrow H(2\pi r) = I'$$

$$\Rightarrow H(2\pi r) = I \left[\frac{r^2}{\rho^2} \right]$$

$$\Rightarrow H = r \left[\frac{I}{2\pi \rho^2} \right]$$

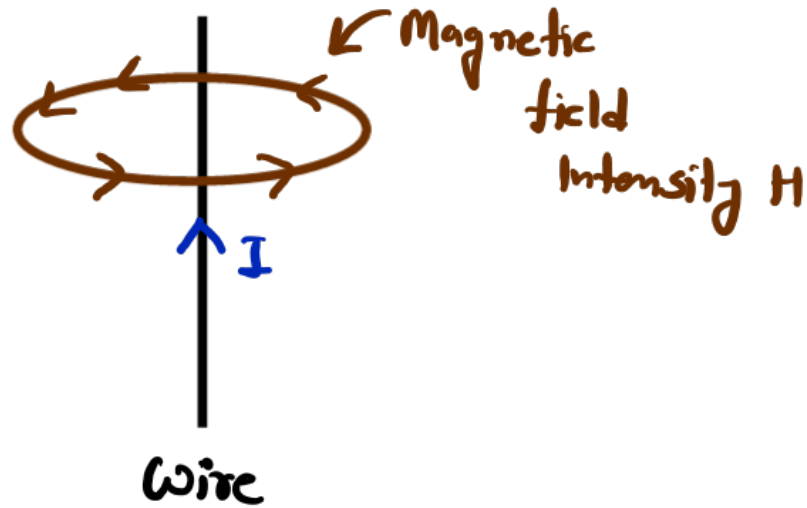
$$\Rightarrow \vec{H} = r \left[\frac{I}{2\pi \rho^2} \right] \hat{i}_\phi$$

Magnetic Field Intensity for Straight Long Wire



Wire

Maxwell's 4th Equation



~> As per Ampere Circuit Law

$$\Rightarrow \oint \vec{H} \cdot d\vec{l} = I \quad \text{--- (1)}$$

~> Total Current $I = I_c + I_d$

Conduction current $\uparrow$ Displacement current
[due to time varying field]

~> If we have current charge density $\vec{J}$

$$\Rightarrow I = \int \vec{J} \cdot d\vec{S} \quad \text{--- (2)}$$

$$\Rightarrow \vec{J} = \vec{J}_c + \vec{J}_d$$

Conduction current charge density $\uparrow$ displacement current charge density [due to time varying field]

→ From eq.ⁿ ① and ②.

$$\Rightarrow \oint \vec{H} \cdot d\vec{l} = \int \vec{J} \cdot d\vec{S} \text{ --- (A)} \quad \Bigg| \quad \boxed{\rightarrow \oint \vec{H} \cdot d\vec{l} = \int (\vec{J}_c + \vec{J}_d) \cdot d\vec{S}} \text{ --- (C)}$$

→ eq.ⁿ (A) is Integral form of Maxwell's 4th eq.ⁿ

→ As per Stokes theorem.

$$\Rightarrow \oint \vec{H} \cdot d\vec{l} = \int (\vec{\nabla} \times \vec{H}) \cdot d\vec{S} = \int \vec{J} \cdot d\vec{S}$$

$$\Rightarrow \vec{\nabla} \times \vec{H} = \vec{J} \text{ --- (B)} \quad \Bigg| \quad \boxed{\rightarrow \vec{\nabla} \times \vec{H} = \vec{J}_c + \vec{J}_d} \text{ --- (D)}$$

→ eq.ⁿ (B) is point form of Maxwell's 4th eq.ⁿ

→ eq.ⁿ (C) & (D) are Maxwell's 4th eq.ⁿ for time varying field.

Magnetic Field (B) and Magnetic Field Intensity (H)

→ Magnetic Field B and Magnetic Field Intensity H is related as per:

$$\Rightarrow \vec{B} \propto \vec{H}$$

$$\Rightarrow \vec{B} = \mu \vec{H}$$

$$\Rightarrow \vec{B} = \mu_0 \mu_r \vec{H}$$

→ where, μ_0 = Absolute permeability = $4\pi \times 10^{-7} \text{ SI}$
 μ_r = Relative permeability

→ Direction of Magnetic field $\vec{B}$ and Magnetic Field Intensity $\vec{H}$ is same.

$$\text{Units} \left\{ \begin{array}{l} \rightarrow H \rightarrow \text{A/m} \\ \rightarrow B \rightarrow \text{Tesla (T)}, \frac{\text{N}}{\text{Am}}, \frac{\text{Wb}}{\text{m}^2}, \text{ Gauss (G)} \end{array} \right.$$

→ If charge q is moving with velocity $\vec{v}$ under magnetic field $\vec{B}$, then Force on charge q is given by:

$$\Rightarrow \vec{F} = q (\vec{v} \times \vec{B})$$

$$\Rightarrow N = (A \cancel{\text{se}}) \left(\frac{\text{m}}{\cancel{\text{se}}} \right) (T)$$

$$\Rightarrow \boxed{T = \frac{N}{Am}}$$

↪ Magnetic Flux ϕ_m

$$\Rightarrow \phi_m = \int \vec{B} \cdot d\vec{S}$$

$$\Rightarrow \text{wb} = (T)(m^2)$$

$$\Rightarrow \boxed{T = \frac{\text{wb}}{m^2}}$$

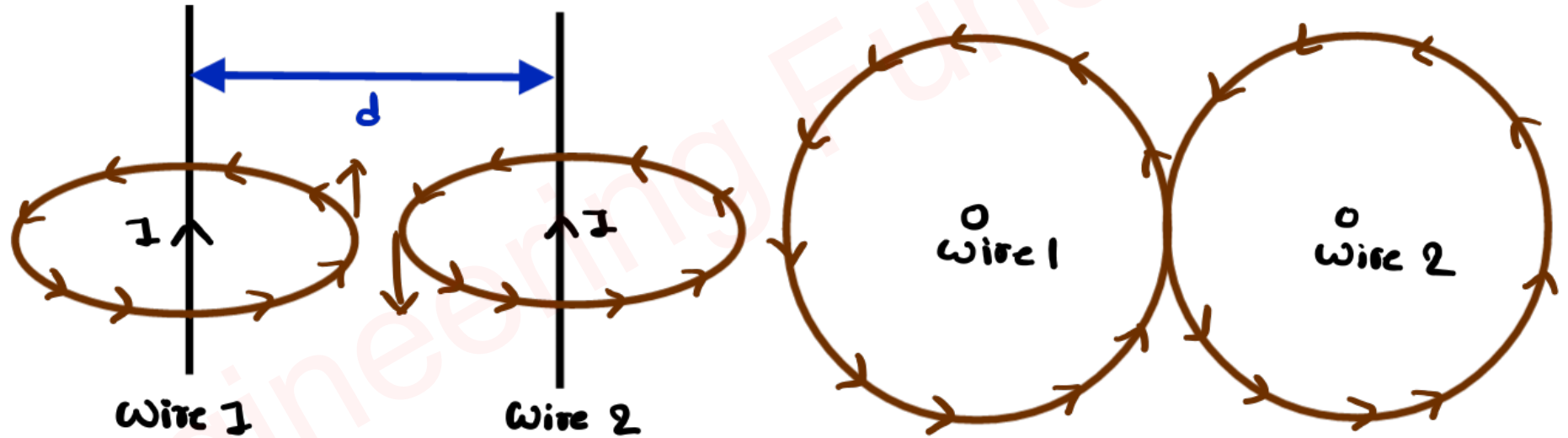
↪ Gauss (G) is unit of Magnetic field B in CGS.

$$\Rightarrow \boxed{1T = 10000 \text{ G}}$$

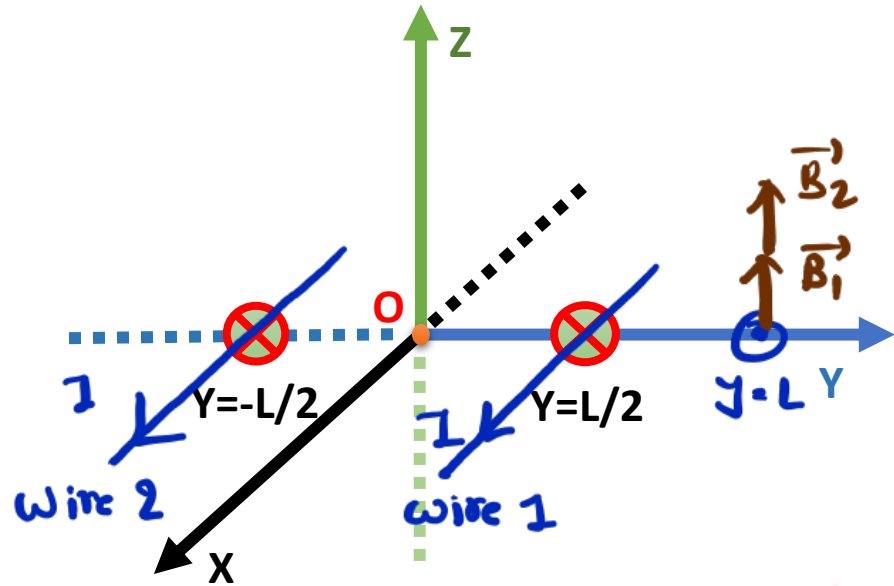
Examples of Magnetic Field

Question 1 – Two Parallel wires separated by a distance d are carrying a current I in the same direction. The magnetic field along a line running parallel to these wires and midway between them:

- a. Depends on I
- ✓ b. Zero
- c. Depends on d
- d. Depends on μ_r



Question 2 – A steady current I is flowing in the x direction through each two infinitely long wires at $y = \pm \frac{L}{2}$, as shown in the figure. The Permeability of the medium is μ_0 . The Magnetic Field $\vec{B}$ at $(0, L, 0)$ is



⇒ Magnetic field Intensity H

$$\Rightarrow H = \frac{I}{2\pi d}$$

⇒ Magnetic field B

$$\Rightarrow B = \frac{\mu_0 I}{2\pi d}$$

⇒ Total Magnetic field

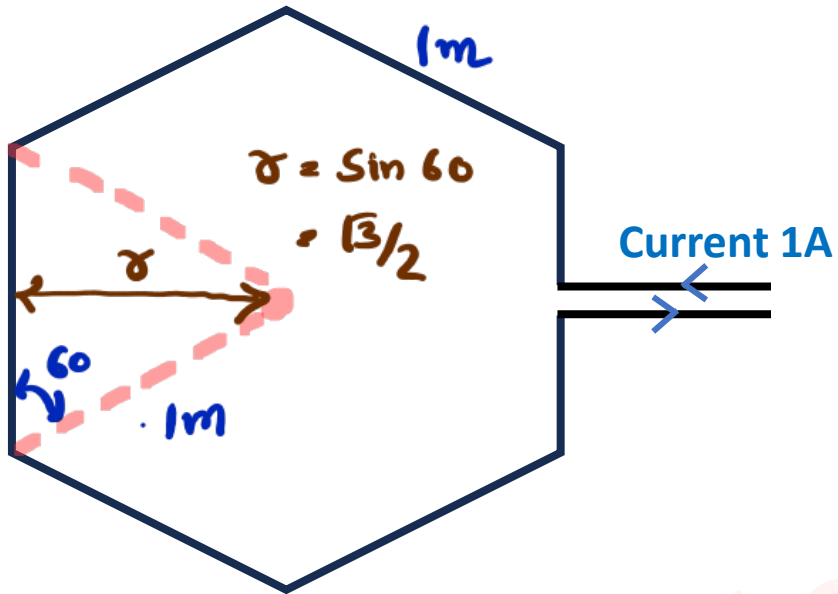
$$\Rightarrow \vec{B} = \vec{B}_1 + \vec{B}_2$$

$$= \frac{\mu_0 I}{2\pi \left(\frac{L}{2}\right)} \hat{i}_2 + \frac{\mu_0 I}{2\pi \left(\frac{3L}{2}\right)} \hat{i}_2$$

$$\Rightarrow \vec{B} = \frac{\mu_0 I}{\pi L} \left[1 + \frac{1}{3} \right] \hat{i}_2$$

$$\Rightarrow \vec{B} = \frac{4\mu_0 I}{3\pi L} \hat{i}_2$$

Question 3 – The magnitude of magnetic flux density B in (μT), at the center of the loop of wire wound as a regular hexagon of side length 1m carrying current 1A and placed in a vacuum as shown in the figure is



~ Magnetic field for polygon is given by

$$\Rightarrow B = \frac{\mu_0 n I}{2\pi r} \sin\left(\frac{\pi}{n}\right)$$

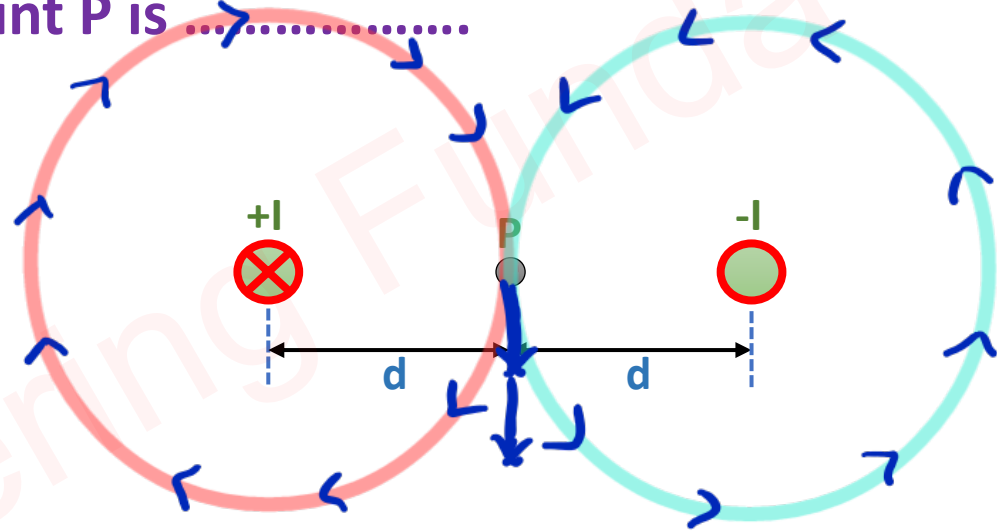
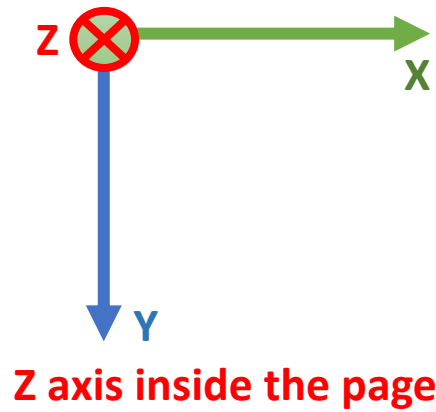
where, n = no of sides of polygon.
 r = 1st distⁿ w.r.t center.

$$\begin{aligned} \Rightarrow B &= \frac{2 \times 10^{-7} \times 6 \times 1}{\pi (\sqrt{3}/2)} \sin\left(\frac{\pi}{6}\right) \\ &= \frac{12 \times 10^{-7}}{\sqrt{3}} \\ &= 6.928 \times 10^{-7} = 0.6928 \times 10^{-6} T \\ &= 0.6928 \mu T \end{aligned}$$

Examples of Magnetic Field Intensity

Question 1 – Two conductors are carrying forward and return current $+I$ and $-I$ as shown in figure. The magnetic field intensity $\vec{H}$ at point P is

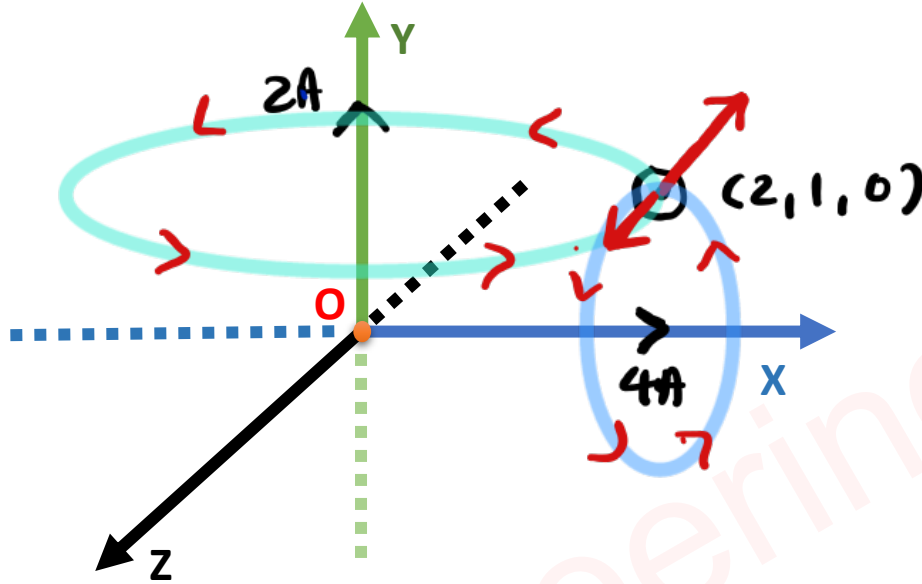
- a. $+\frac{I}{\pi d} \hat{i}_y$
- b. $+\frac{I}{2\pi d} \hat{i}_y$
- c. $-\frac{I}{\pi d} \hat{i}_y$
- d. $-\frac{I}{2\pi d} \hat{i}_y$



$$\begin{aligned}\vec{H} &= \vec{H}_1 + \vec{H}_2 \\ &= \frac{I}{2\pi d} \hat{i}_y + \frac{I}{2\pi d} \hat{i}_y \\ &= \frac{I}{\pi d} \hat{i}_y\end{aligned}$$

Question 2 – An infinitely extended wire is laid along x axis and is carrying current of 4A in the +ve X direction. Another infinitely extended wire is laid along the y axis and is carrying 2A current in +ve Y direction. Magnetic field intensity at (2, 1, 0) will be

- a. $+\frac{3}{2\pi} \hat{k} \frac{\text{Wb}}{\text{m}^2}$
- b. $+\frac{4}{3\pi} \hat{k} \frac{\text{A}}{\text{m}}$
- ☒ c. $+\frac{3}{2\pi} \hat{k} \frac{\text{A}}{\text{m}}$
- d. $0 \frac{\text{A}}{\text{m}}$



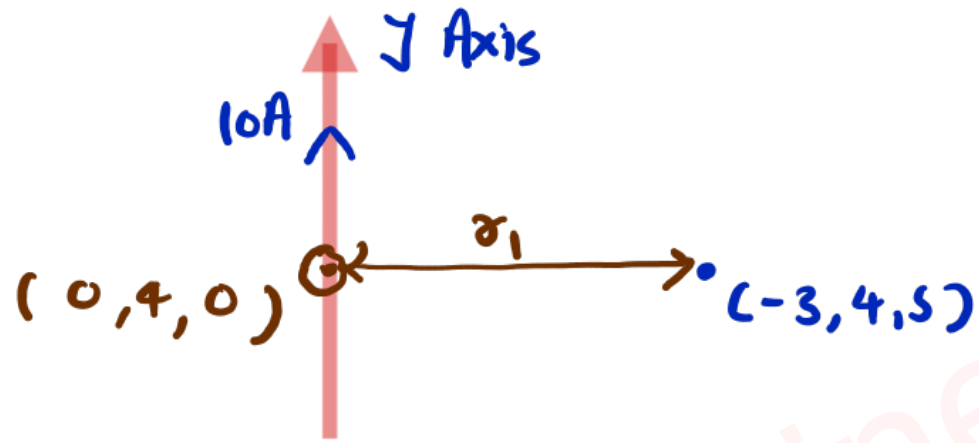
$$\begin{aligned} \rightarrow \vec{H}_1 &= \frac{I_1}{2\pi d_1} (-\hat{k}) \\ &= \frac{2}{2\pi (2)} (-\hat{k}) \\ &= \frac{-1}{2\pi} \hat{k} \end{aligned}$$

$$\begin{aligned} \sim \vec{H}_2 &= \frac{I_2}{2\pi d_2} \hat{k} \\ &= \frac{4}{2\pi (1)} \hat{k} \\ &= \frac{2}{\pi} \hat{k} \end{aligned}$$

$$\begin{aligned} \sim \vec{H} &= \vec{H}_1 + \vec{H}_2 \\ &= \frac{1}{\pi} \left[2 - \frac{1}{2} \right] \hat{k} \\ &= \frac{3}{2\pi} \hat{k} \left(\frac{\text{A}}{\text{m}} \right) \end{aligned}$$

Example of Magnetic Field Intensity

Question 3 – Y and Z-Axis carrying current by 10A and -20A, respectively. Find the $\vec{H}$ at $(-3, 4, 5)\text{m}$.



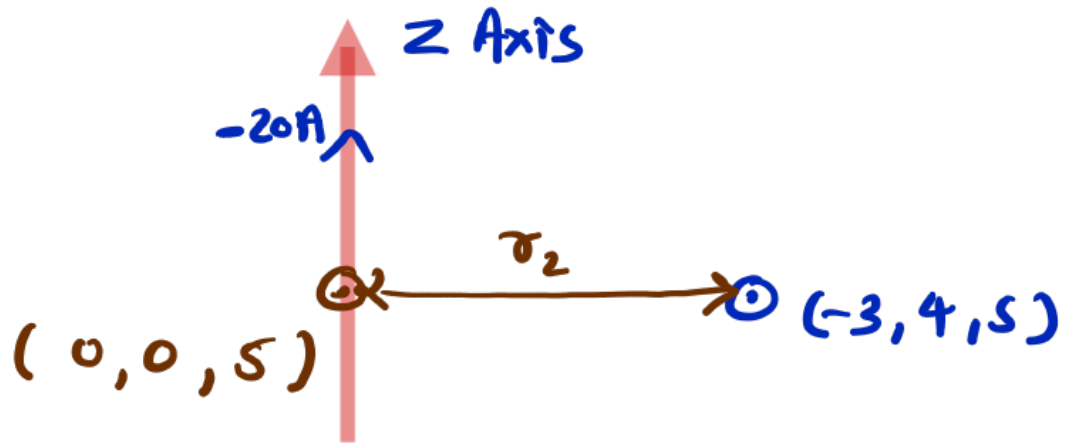
$$\begin{aligned}\rightarrow \vec{r}_1 &= (-3, 4, 5) - (0, 4, 0) \\ &= (-3, 0, 5) \text{ m}\end{aligned}$$

$$\begin{aligned}\rightarrow r_1 &= \sqrt{(-3)^2 + 0^2 + 5^2} \\ &= \sqrt{34} \text{ m}\end{aligned}$$

$$\begin{matrix} \xrightarrow{\quad} \\ x & y & z & x & y & z \end{matrix}$$

$$\begin{aligned}\rightarrow \hat{r}_{1,y} &= \hat{a}_y \times \hat{r}_1 \\ &= \hat{a}_y \times (-3\hat{a}_x + 5\hat{a}_z) \\ &= \frac{3\hat{a}_z}{\sqrt{34}} + \frac{5\hat{a}_x}{\sqrt{34}}\end{aligned}$$

$$\begin{aligned}\rightarrow \vec{H}_1 &= \frac{I_1}{2\pi r_1} \hat{r}_{1,y} \\ &= \frac{10}{2\pi \sqrt{34}} \times \left[\frac{3\hat{a}_z + 5\hat{a}_x}{\sqrt{34}} \right] \\ &= (0.234 \hat{a}_x + 0.14 \hat{a}_z) \text{ A/m}\end{aligned}$$



$$\begin{aligned}\rightarrow \vec{H}_2 &= \frac{I_2}{2\pi r_2} \hat{a}_{22} \\ &= \frac{+20}{2\pi 5} \times \left(+\frac{3}{5} \hat{a}_y + \frac{4}{5} \hat{a}_x \right) \\ &= (0.509 \hat{a}_x + 0.382 \hat{a}_y) \text{ A/m}\end{aligned}$$

$$\begin{aligned}\therefore \vec{H} &= \vec{H}_1 + \vec{H}_2 = (0.234 \hat{a}_x + 0.14 \hat{a}_z) + (0.509 \hat{a}_x + 0.382 \hat{a}_y) \\ &= (0.743 \hat{a}_x + 0.382 \hat{a}_y + 0.14 \hat{a}_z) \text{ A/m}\end{aligned}$$

$$\begin{aligned}\rightarrow \vec{r}_2 &= (-3, 4, 5) - (0, 0, 5) \\ &= (-3, 4, 0)\end{aligned}$$

$$\begin{aligned}\rightarrow r_2 &= \sqrt{(-3)^2 + 4^2 + 0^2} \\ &= \sqrt{25} = 5 \text{ m}\end{aligned}$$

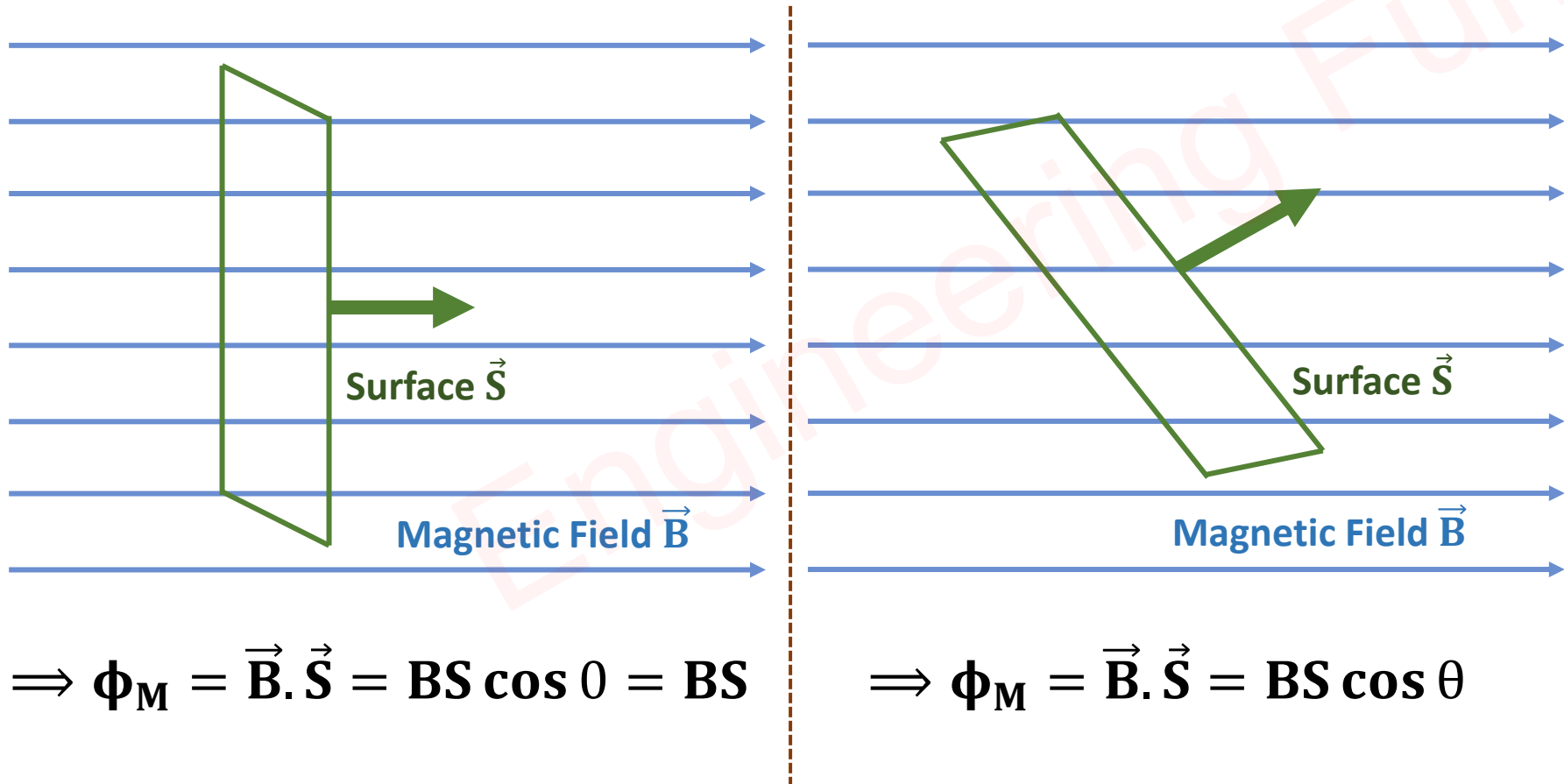
$$\begin{aligned}\rightarrow \hat{a}_{22} &= \hat{a}_2 \times \hat{r}_2 \\ &= \hat{a}_2 \times \frac{(-3\hat{a}_x + 4\hat{a}_y)}{5} \\ &= -\frac{3}{5} \hat{a}_y - \frac{4}{5} \hat{a}_x\end{aligned}$$

$\xrightarrow{\quad}$
x y z x y z

Magnetic Flux

- Magnetic Flux is Amount of Magnetic Field $\vec{B}$ passing through Cross-Sectional Area $\vec{S}$.

$$\Rightarrow \Phi_B = \int \vec{B} \cdot d\vec{S}$$



- Unit of Magnetic Flux is Weber (Wb).

$$\Rightarrow \text{Wb} = (\text{T})\text{m}^2$$

$$\Rightarrow \text{Wb} = \left(\frac{\text{N}}{\text{Am}} \right) \text{m}^2$$

$$\Rightarrow \text{Wb} = \frac{\text{Nm}}{\text{A}}$$

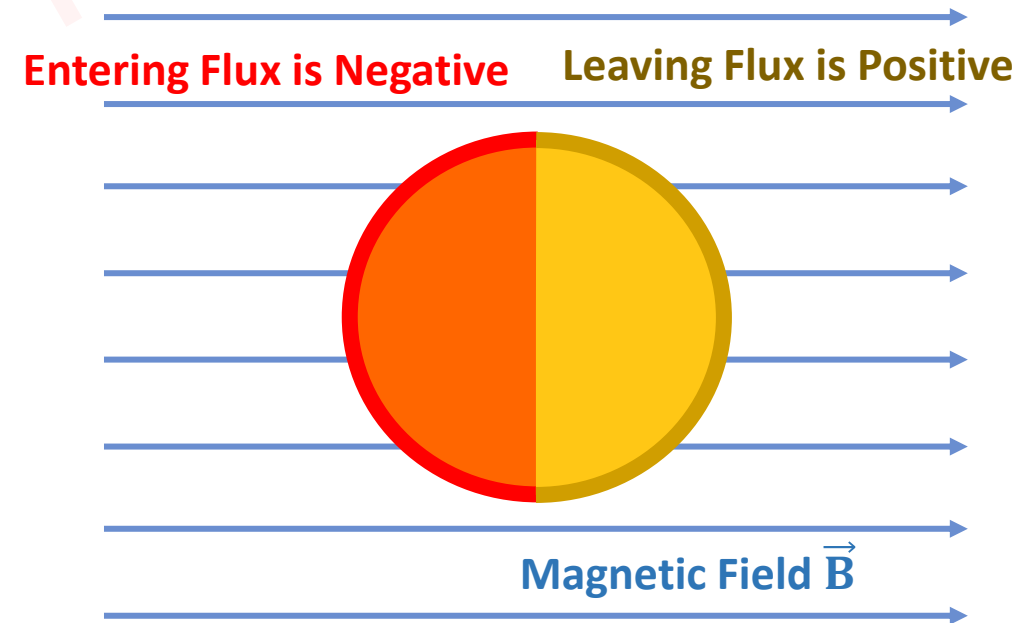
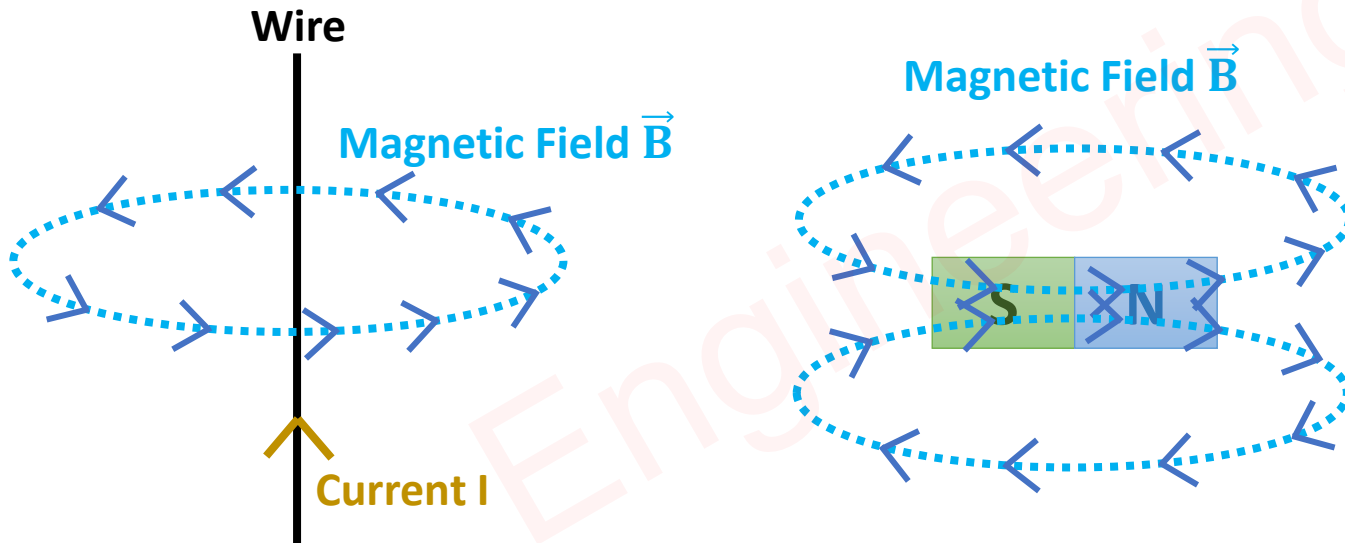
Gauss's Law for Magnetic Field

- Gauss's Law for the Magnetic Field States, "For Closed Surface, Magnetic Flux is Zero."

$$\Rightarrow \Phi_B = \oint \vec{B} \cdot d\vec{S} = 0$$

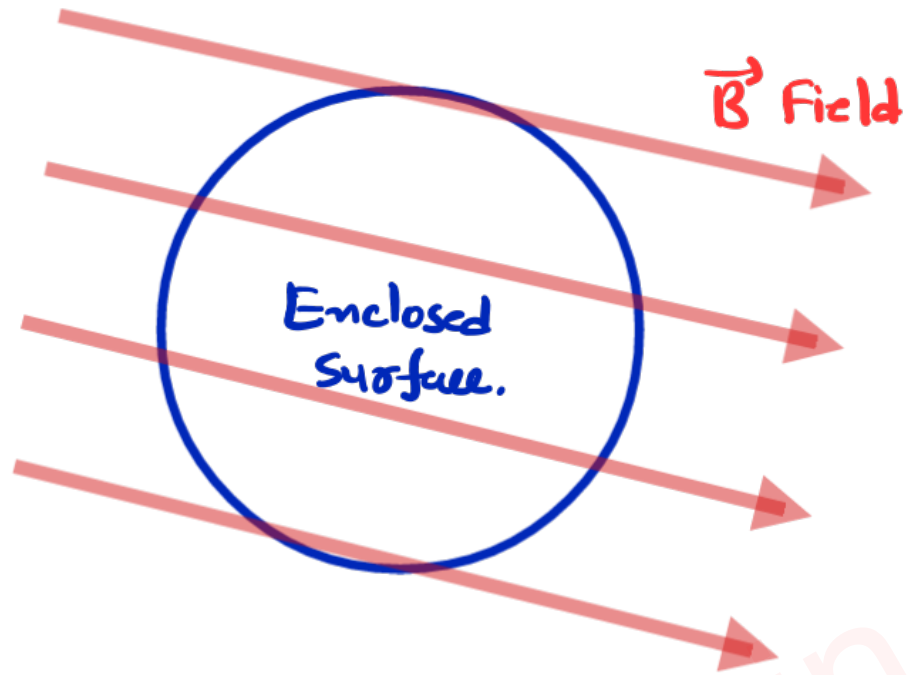
Proof:

- Magnetic Field lines always stay in a closed loop.



$$\Rightarrow \Phi_B = \oint \vec{B} \cdot d\vec{S} = \Phi_E + \Phi_L = 0$$

Maxwell's 2nd Equation



→ As per Gauss Law of Magnetic Field.

$$\Rightarrow \oint \vec{B} \cdot d\vec{S} = 0 \text{ — ①}$$

→ Eq.ⁿ ① is Maxwell's 2nd eq.ⁿ with Integral Form.

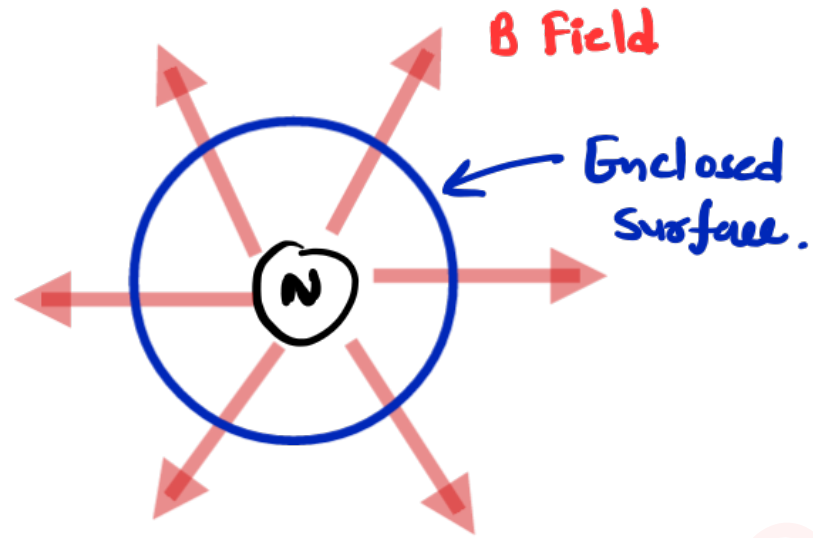
→ As per Divergence Theorem.

$$\Rightarrow \oint \vec{B} \cdot d\vec{S} = \int (\nabla \cdot \vec{B}) dV = 0$$

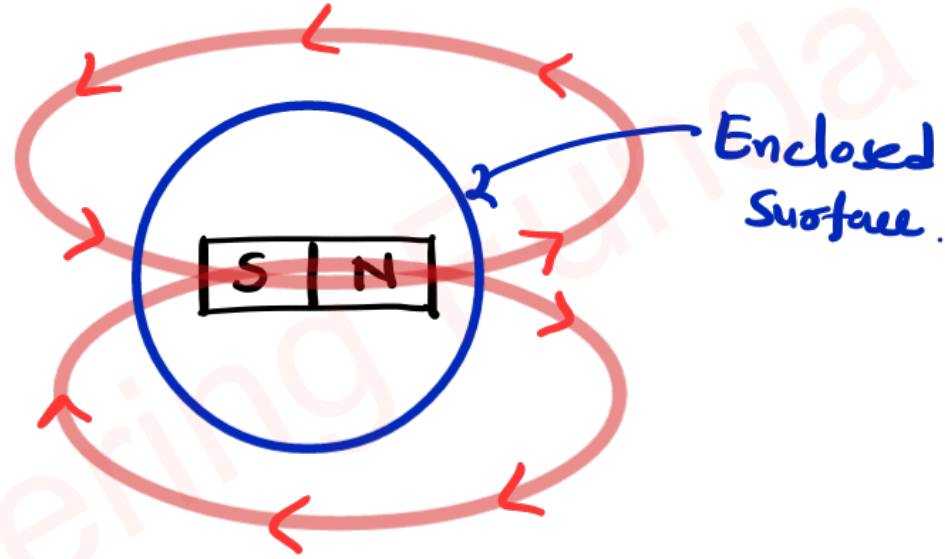
$$\Rightarrow \nabla \cdot \vec{B} = 0 \text{ — ②}$$

→ Eq.ⁿ ② is Maxwell's 2nd eq.ⁿ point / Differential Form.

★ meaning - There is no existence of magnetic monopole.



$$\oint \vec{B} \cdot d\vec{s} \neq 0$$



$$\oint \vec{B} \cdot d\vec{s} = 0$$

Examples of Gauss Law of Magnetic Field

Question 1 – The Flux density at a point in space is given by

$$\vec{B} = 4x\hat{a}_x + 2ky\hat{a}_y + 8\hat{a}_z \text{ W}_b/\text{m}^2$$

The value of constant k must be equal.

- a. ☒ -2
- b. ☐ -0.5
- c. ☐ 0.5
- d. ☐ 2

⇒ As per Maxwell's 2nd eq.ⁿ in point form

$$\Rightarrow \nabla \cdot \vec{B} = 0$$

$$\Rightarrow \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} = 0$$

$$\Rightarrow 4 + 2k + 0 = 0$$

$$\Rightarrow k = -2$$

Question 2 – The following four vector fields are given in a cartesian coordinate system. The vector field does not satisfy the property of magnetic flux density is:

a. $y^2 \mathbf{a}_x + z^2 \mathbf{a}_y + x^2 \mathbf{a}_z$ ✓

b. $z^2 \mathbf{a}_x + x^2 \mathbf{a}_y + y^2 \mathbf{a}_z$ ✓

c. $x^2 \mathbf{a}_x + y^2 \mathbf{a}_y + z^2 \mathbf{a}_z$ ✗

d. $y^2 z^2 \mathbf{a}_x + x^2 z^2 \mathbf{a}_y + x^2 y^2 \mathbf{a}_z$ ✓

⇒ As per Maxwell's 2nd eqⁿ in point Form

$$\Rightarrow \nabla \cdot \vec{B} = 0$$

$$\Rightarrow \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} = 0$$

Examples of Magnetic Vector Potential

Question 1 – The line integral of the vector potential A around the boundary of a surface S represents:

- a. ✓ Flux through the surface S
- b. Flux density in the surface S
- c. Magnetic Density
- d. Current Density

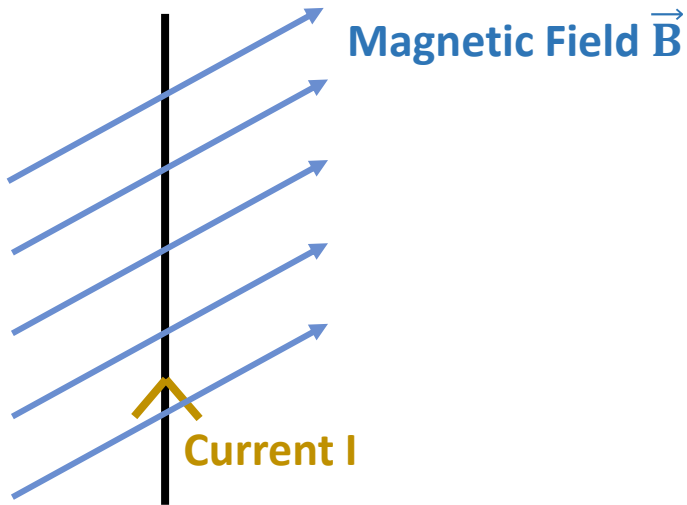
⇒ The relation bet.ⁿ magnetic field $\vec{B}$ & Vector Magnetic Potential $\vec{A}$ is given by:

$$\Rightarrow \vec{B} = \nabla \times \vec{A}$$

$$\begin{aligned} \Rightarrow \int \vec{A} \cdot d\vec{\ell} &= \int (\nabla \times \vec{A}) \cdot d\vec{S} \quad [\text{Stoke's Theorem}] \\ &= \int \vec{B} \cdot d\vec{S} \\ &= \Phi_B \quad [\text{Mag. Flux}] \end{aligned}$$

Question 2 – The magnetic vector potential in a region is defined by $\vec{A} = e^{-y} \sin x \mathbf{a}_z$. An infinitely long conductor, having a cross-section area 5mm^2 and carrying a DC current 5A in the y direction, passes through this region as shown in figure. Determine the expression for:

- Magnetic field
- Force per unit length



$$\Rightarrow \vec{B} = \nabla \times \vec{A}$$

$$= \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & e^{-y} \sin x \end{vmatrix}$$

$$= \mathbf{a}_x \left(\frac{\partial}{\partial y} (e^{-y} \sin x) \right) - \mathbf{a}_y \left(\frac{\partial}{\partial x} (e^{-y} \sin x) \right)$$

$$\boxed{\vec{B} = -e^{-y} \sin x \mathbf{a}_x - e^{-y} \cos x \mathbf{a}_y}$$

$\Rightarrow$ For current carrying wire, Force.

$$\Rightarrow \vec{F} = I d\vec{l} \times \vec{B}$$

$$\Rightarrow \vec{F} = 5 (dl \hat{a}_y) \times [-\hat{e}^{-j} \sin x \hat{a}_x - \hat{e}^{-j} \cos x \hat{a}_y]$$

$\begin{matrix} x & y & z & x & y & z \\ \hline & & & & & \end{matrix}$

$$\Rightarrow \frac{\vec{F}}{dl} = 5 \hat{a}_y \times [-\hat{e}^{-j} \sin x \hat{a}_x - \hat{e}^{-j} \cos x \hat{a}_y]$$

$$= 5 (-\hat{e}^{-j} \sin x) (-\hat{a}_z)$$

$$= 5 \hat{e}^{-j} \sin x \hat{a}_z$$

Faraday's Law

Outlines

- Basics of Faraday's Law
- Maxwell's 3rd Equation for Time-Varying Field
- Applications of Faraday's Law

Engineering Funda

Faraday's Law

Basics of Faraday's Law

- In the Static Magnetic Field, the Circuit has no induced current.
- In a Time-Varying Magnetic Field, There is an induced current in the circuit.

☐ Statement of Faraday's Law

☐ "Rate of change of magnetic flux with respect to time generates induced emf in the circuit."

$$\Rightarrow V_{\text{emf}} = - \frac{d\Phi_B}{dt}$$

- Where (-Ve) Sign is due to Induced emf and the time-varying field is in the opposite direction.

Maxwell's 3rd Equation for Time-Varying Field

- As per Faraday's Law,

$$\Rightarrow V_{\text{emf}} = - \frac{d\Phi_B}{dt}$$

[$\because$ (-Ve) Sign is due to Induced emf and the time-varying field is in the opposite direction.]

- As per Potential,

$$\Rightarrow V_{\text{emf}} = + \int \vec{E} \cdot d\vec{l}$$

[$\because$ (+Ve) Sign is due to work done on the charges.]

- From above equation,

$$\Rightarrow \int \vec{E} \cdot d\vec{l} = - \frac{d\Phi_B}{dt}$$

$$\Rightarrow \int \vec{E} \cdot d\vec{l} = - \frac{d \int \vec{B} \cdot d\vec{S}}{dt}$$

[$\because$ Magnetic Flux $\Phi_B = \int \vec{B} \cdot d\vec{S}$]

$$\Rightarrow \int (\nabla \times \vec{E}) \cdot d\vec{S} = - \frac{d \int \vec{B} \cdot d\vec{S}}{dt}$$

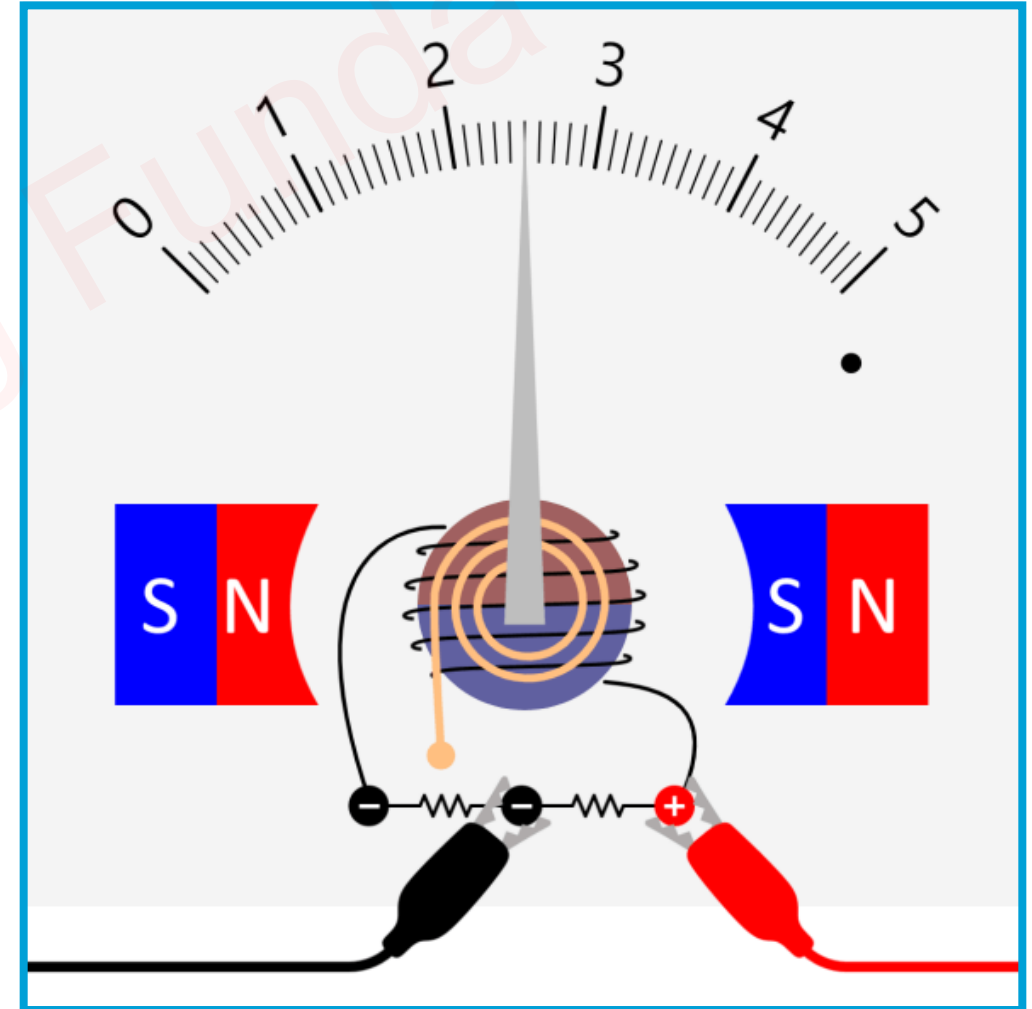
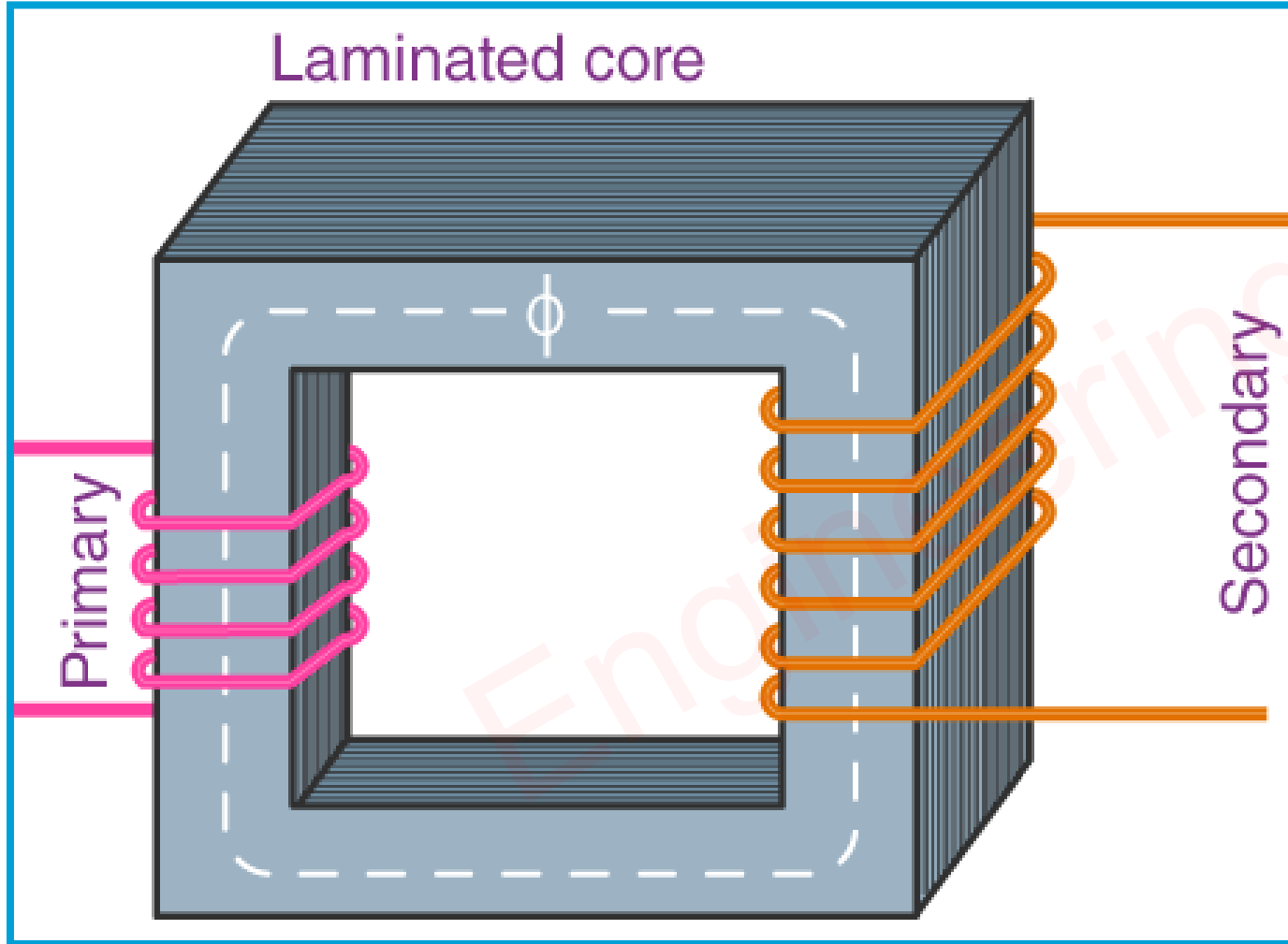
[$\because$ Stocks Theorem, $\int \vec{E} \cdot d\vec{l} = \int (\nabla \times \vec{E}) \cdot d\vec{S}$]

$$\Rightarrow \nabla \times \vec{E} = - \frac{d\vec{B}}{dt}$$

- Above equations are Maxwell's 3rd equation in Integral and Differential Form.

Applications of Faraday's Law

- Transformer
- Ammeter



Examples of Faraday's Law

Question 1 – The Laws of electromagnetic induction [Faraday's and Lenz's Law] are summarized in following equation:

a. $e = iR$

b. $e = L \frac{di}{dt}$

☒ c. $e = - \frac{d\Phi}{dt}$

d. None of Above

∴ Faraday's Law

$$\Rightarrow V_{emf} = \frac{d\Phi_B}{dt}$$

∴ Lenz's Law.

$$\Rightarrow V_{emf} = - \frac{d\Phi_B}{dt}$$

Question 2 – Static Magnetic fields induce currents in closed conducting loops. [True/False] ✓

~> Faraday's Law

$$\Rightarrow V_{\text{emf}} = - \frac{d\Phi_B}{dt}$$

$$\Rightarrow V_{\text{emf}} = - \frac{d(BA)}{dt}$$

$$\Rightarrow V_{\text{emf}} = -A \frac{dB}{dt}$$

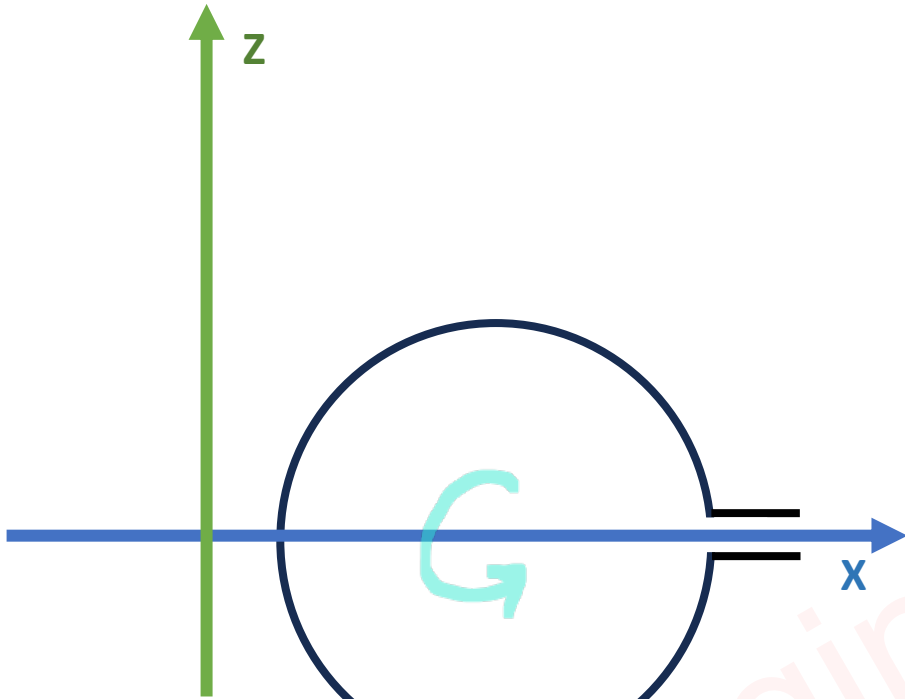
$$\Rightarrow V_{\text{emf}} \propto \frac{dB}{dt}$$

Question 3 – An electromagnetic field is radiated from

- a. A stationary point charges [E Field]
- b. A capacitor with DC voltage [E Field]
- c. A capacitor carrier DC current [B Field]
- ✓ d. An oscillating dipole

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Question 4 – A circular turn of radius 1m revolves at 60 rpm about its diameter aligned with the x axis as shown in the figure. The value of $\mu_0 = 4\pi \times 10^{-7}\text{ SI}$. If a uniform magnetic field intensity $\vec{H} = 10^7 \text{ a}_z \text{ A/m}$ is applied. Then the peak value of induced voltage is 248.05 Volt



$$\sim V_{\text{emf}} = -\frac{d\phi_m}{dt} = -\frac{d(BA)}{dt} = -\frac{d(\mu_0 H A)}{dt} = -\mu_0 H \frac{dA}{dt}$$

$$\sim A_{\text{max}} = \pi r^2 = \pi \text{ m}^2$$

$$A = A_{\text{max}} \sin(\omega t + \phi) \\ = \pi \sin(2\pi t + \phi)$$

$$\begin{aligned} \rightarrow \omega &= 60 \text{ rpm} \\ &= \cancel{60} \frac{(2\pi)}{\cancel{60}} \\ &= 2\pi \text{ rad/sec} \end{aligned}$$

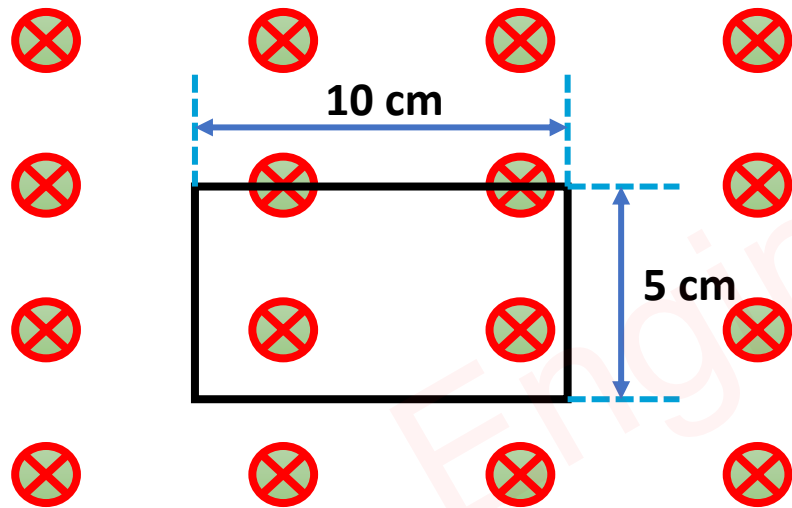
$$\sim V_{\text{emf}} = -\mu_0 H \left[\pi (2\pi \omega \cos(2\pi t + \phi)) \right]$$

$$\begin{aligned} &= -4\pi \times \cancel{10^{-7}} \times \cancel{10^7} \times \pi \times (2\pi) \cos(2\pi t + \phi) \\ &= -8\pi^3 \cos(2\pi t + \phi) \end{aligned}$$

$$V_{\text{max}} = 8\pi^3 = 248.05 \text{ Volt}$$

Examples of Faraday's Law

Question 5 – Consider a one turn rectangular loop of wire placed in a uniform magnetic field as shown in the figure. The plane of the loop is perpendicular to the field lines. The resistance of the loop is 0.4Ω and its inductance is negligible. The magnetic flux density (in Tesla) is a function of time, $B(t) = 0.25 \sin \omega t$. Where, $\omega = 100\pi$ Rad/sec. The power absorbed by the loop from the magnetic field is



$$\leadsto P_{\text{absorbed}} = \frac{V_{\text{rms}}^2}{R}, \quad V_{\text{rms}} = \frac{V_m}{\sqrt{2}}$$

$$\begin{aligned} \leadsto V_{\text{emf}} &= - \frac{d\phi_m}{dt} \\ &= - \frac{d(BA)}{dt} \end{aligned}$$

$$= - 50 \times 10^{-4} \frac{d(0.25 \sin 100\pi t)}{dt}$$

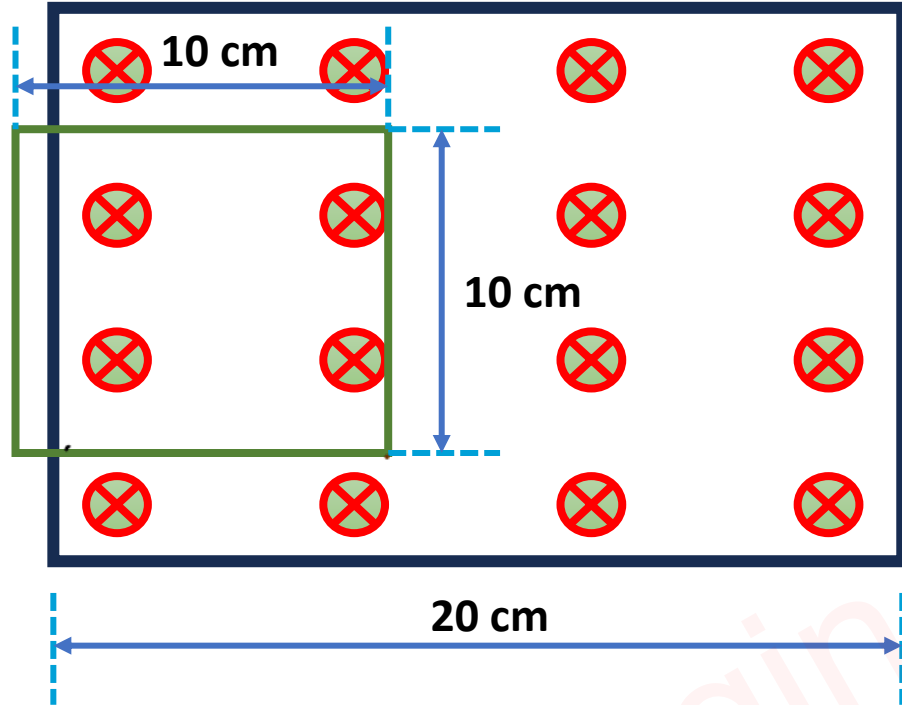
$$= - 50 \times 10^{-4} \times 0.25 \times 100\pi \cos(100\pi t)$$

$$= \underline{-0.3925 \cos(100\pi t)}$$

$$\leadsto V_{\text{rms}} = \frac{V_m}{\sqrt{2}} = \frac{0.3925}{\sqrt{2}} = 0.2775 \text{ Volt}$$

$$\leadsto P_{\text{absorbed}} = \frac{V_{\text{rms}}^2}{R} = \frac{0.2775^2}{0.4} = 0.192 \text{ W}$$

Question 6 – A square coil of 10cm side is moved through a steady magnetic field of $1 \text{ W}_b/\text{m}^2$ at a constant velocity of 2m/sec with its plane perpendicular to the field as shown in figure. Plot the variation of induced emf as the coil moves along the field.



→ Case-I - Coil Enters mag. field.

$$0 < t < 0.05 \text{ sec}$$

$$- A = d_1 d_2 = 0.1 (2t) = 0.2t \text{ m}^2$$

$$- V_{\text{emf}} = - \frac{d\phi}{dt} = - \frac{d(BA)}{dt} = -0.2 \text{ Volt}$$

→ Case-II - Coil is inside mag. field

$$0.05 < t < 0.1 \text{ sec}$$

$$- A = 0.1^2 \text{ m}^2 = 0.01 \text{ m}^2$$

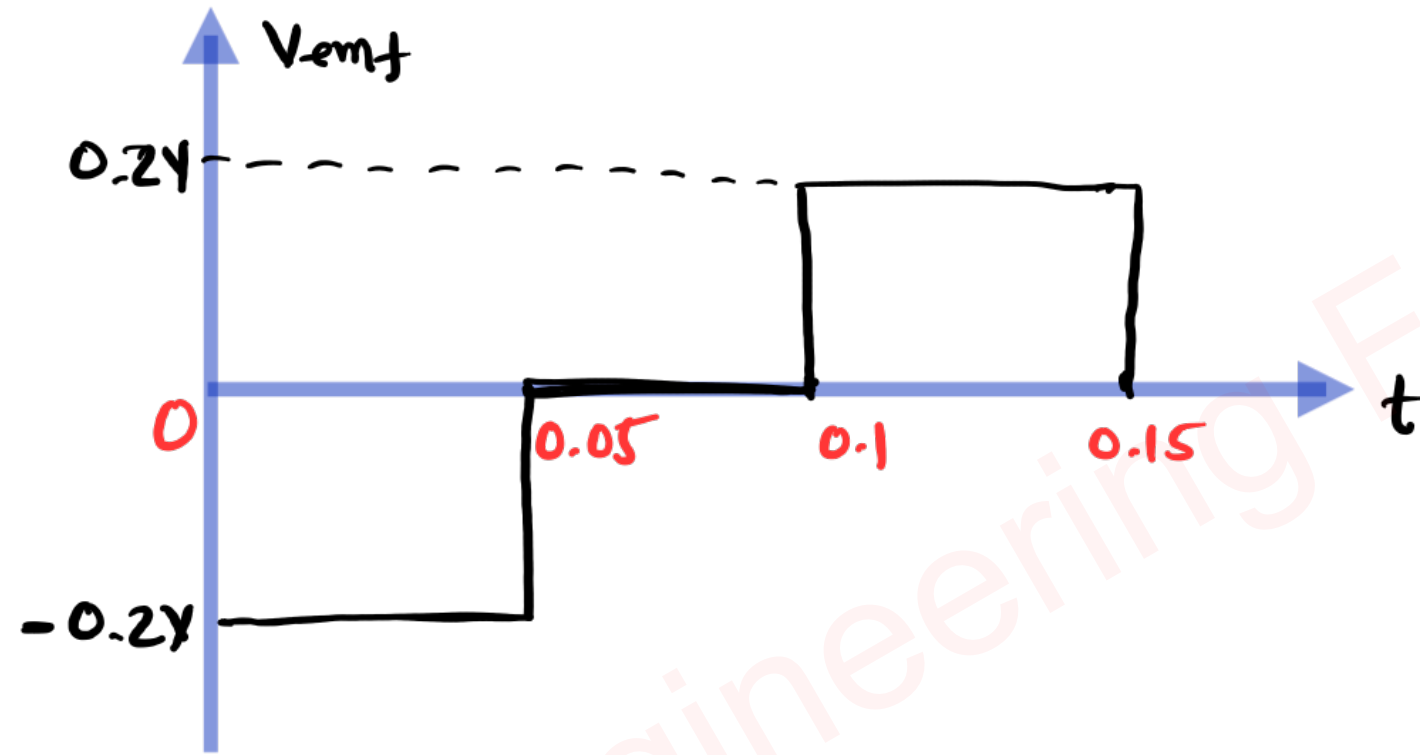
$$- V_{\text{emf}} = - \frac{d\phi}{dt} = 0$$

→ Case-III Coil Leaves mag. field

$$0.1 < t < 0.15 \text{ sec}$$

$$\begin{aligned} - A &= d_1 d_2 = 0.1 (0.1 - 2t) \\ &= 0.01 - 0.2t \text{ m}^2 \end{aligned}$$

$$-V_{emf} = -\frac{d\phi}{dt} = -\frac{d(BA)}{dt} = 0.2 \text{ Volt}$$



Maxwell Equations

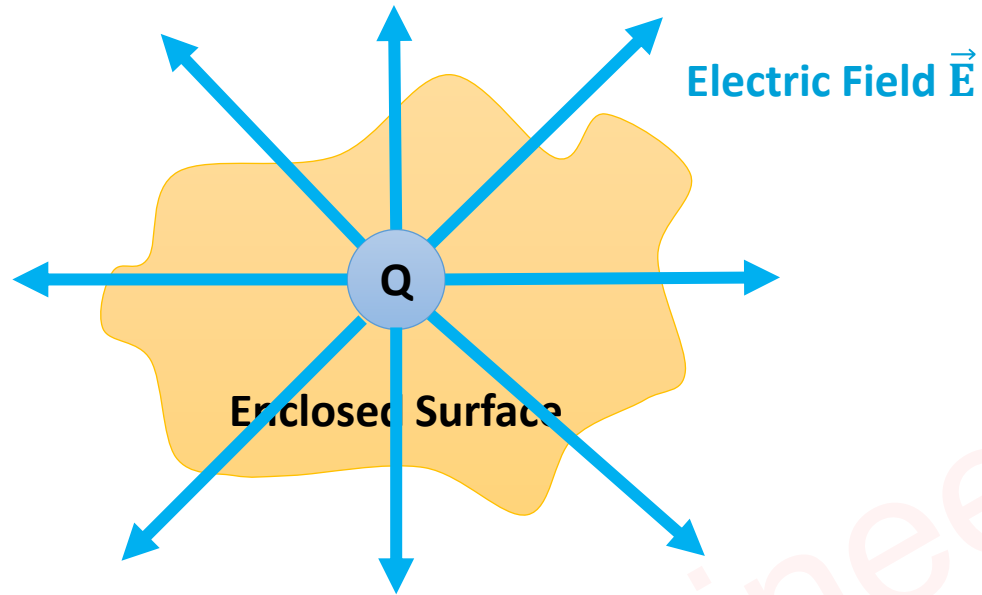
Outlines

- 1st Equation: Gauss Law for Electric Field
- 2nd Equation: Gauss Law for Magnetic Field
- 3rd Equation: Faraday's Law and Potential
- 4th Equation: Ampere Circuit Law

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Maxwell's 1st Equation {Gauss Law for Electric Field}

- Gauss Law for Electric Fields explains the existence of Electric Flux through Closed surface possible only if there is a charge enclosed by the closed surface.



- Integral Form:**

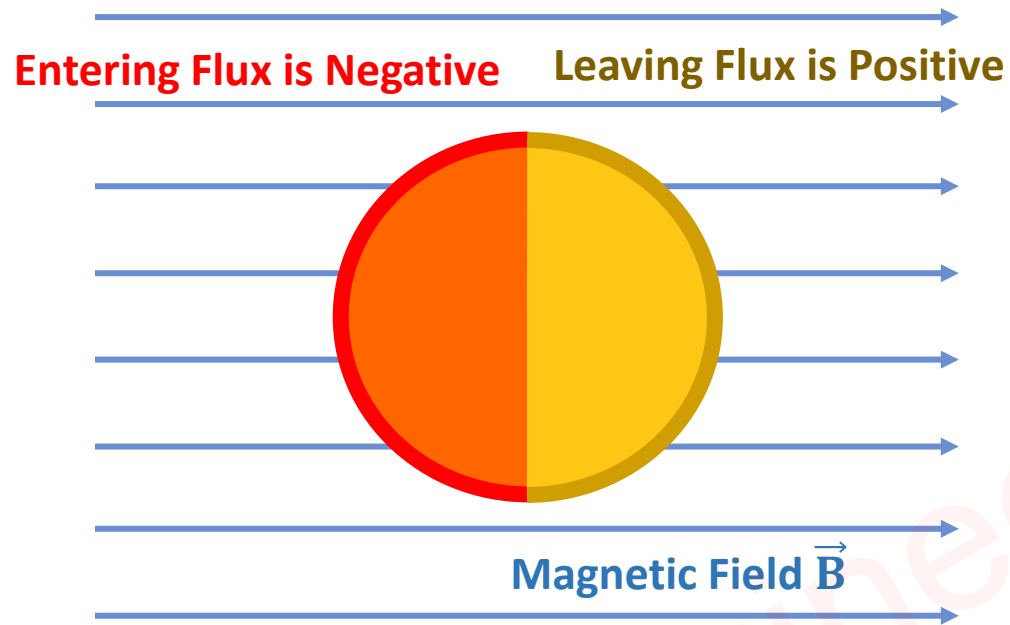
$$\Rightarrow \oint \vec{D} \cdot d\vec{S} = Q \quad \text{Or} \quad \Rightarrow \oint \vec{E} \cdot d\vec{S} = Q/\epsilon_0$$

- Differential/Point Form:**

$$\Rightarrow \nabla \cdot \vec{D} = \rho_V \quad \text{Or} \quad \Rightarrow \nabla \cdot \vec{E} = \rho_V/\epsilon_0$$

Maxwell's 2nd Equation {Gauss Law for Magnetic Field}

- Gauss Law for Magnetic Fields explains that the Magnetic Flux through Closed surface is zero. It also explains that there is no existence of a Magnetic Monopole.



- Integral Form:

$$\Rightarrow \oint \vec{B} \cdot d\vec{S} = 0$$

- Differential/Point Form:

$$\Rightarrow \nabla \cdot \vec{B} = 0$$

Maxwell's 3rd Equation {Faraday's Law and Potential}

- For a Non-Time Varying Field, Potential in a Closed Path is Zero.

- Integral Form:

$$\Rightarrow \oint \vec{E} \cdot d\vec{l} = 0$$

- Differential/Point Form:

$$\Rightarrow \nabla \times \vec{E} = 0$$

- For a Time-Varying Field, Change in the Magnetic Field generates an Electric Field.

- Integral Form:

$$\Rightarrow \oint \vec{E} \cdot d\vec{l} = - \frac{d \int \vec{B} \cdot d\vec{S}}{dt}$$

- Differential/Point Form:

$$\Rightarrow \nabla \times \vec{E} = - \frac{d\vec{B}}{dt}$$

Maxwell's 4th Equation {Ampere Circuit Law}

- Magnetic Field in closed path generates Electric Current and Change in Electric Field also generates Electric Current in a wire.
- Integral Form:

$$\Rightarrow \oint \vec{H} \cdot d\vec{l} = I_c + I_d = \int (\vec{J}_c + \vec{J}_d) \cdot d\vec{S}$$

- Differential Form:

$$\Rightarrow \nabla \times \vec{H} = \vec{J}_c + \vec{J}_d$$

- Where,

$$\Rightarrow \vec{J}_d = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

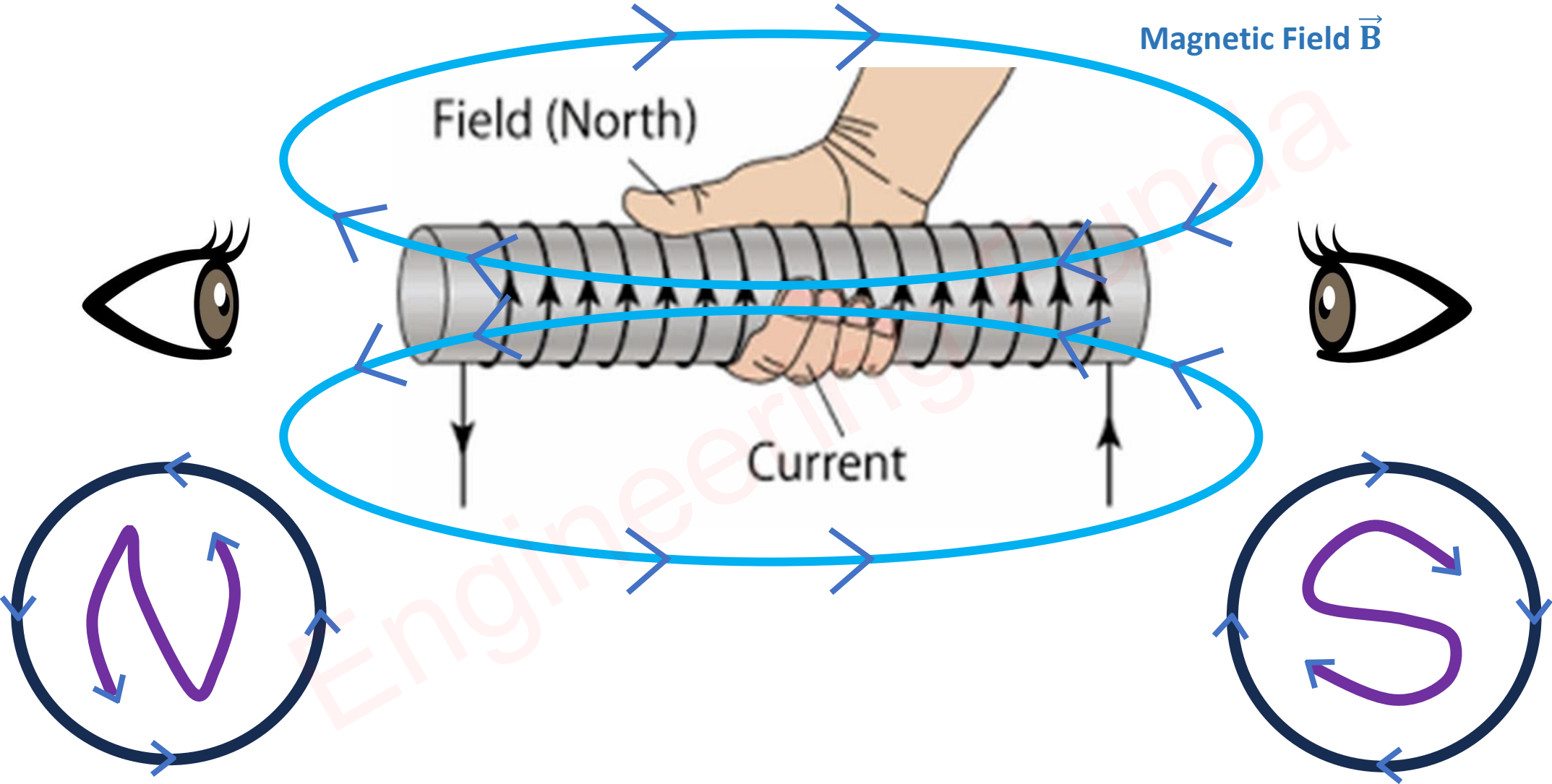
Solenoid

Outlines

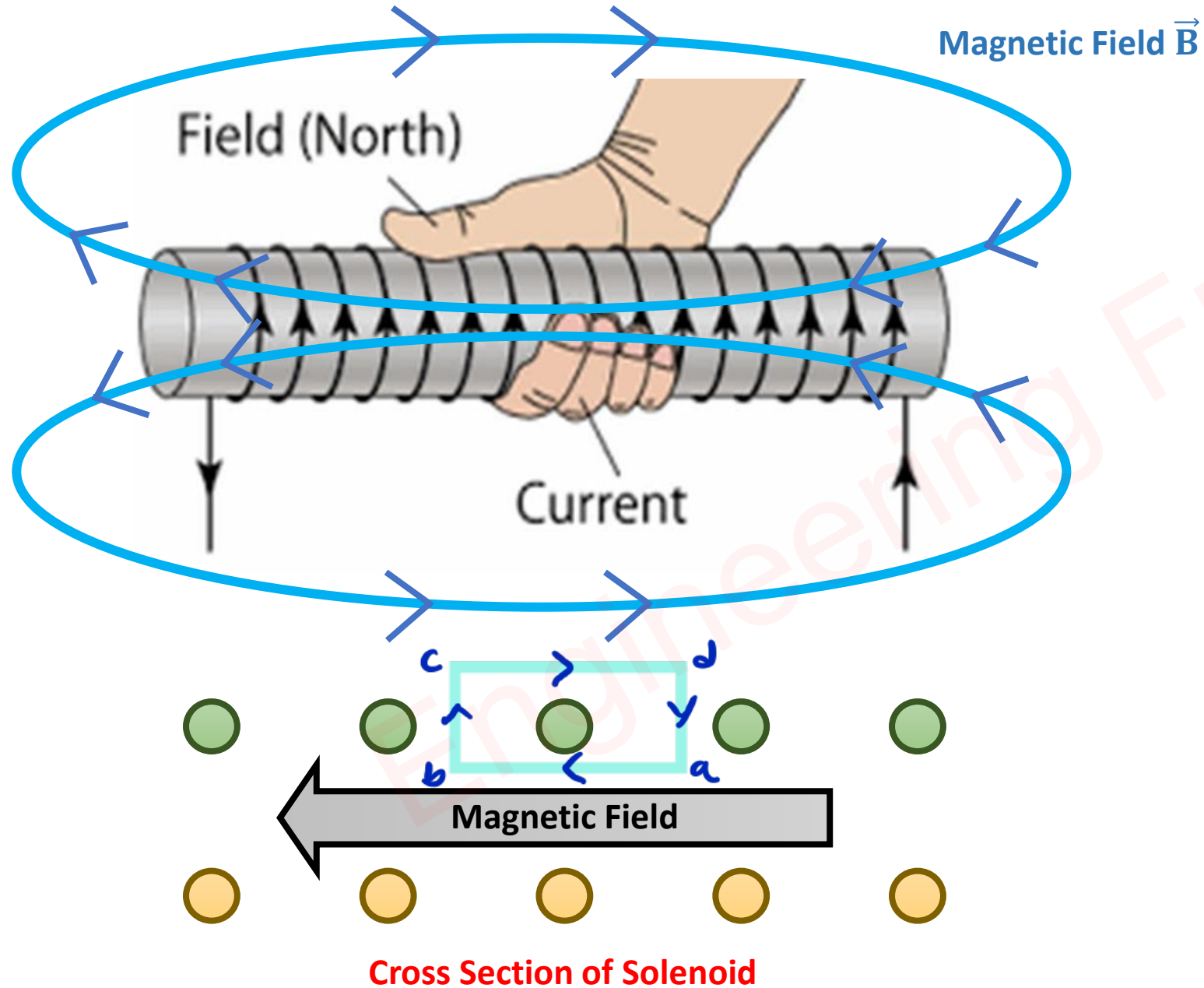
- Structure of Solenoid
- Generation of Magnetic Field in Solenoid
- Direction of Magnetic Field in Solenoid
- Derivation of Magnetic Field in Solenoid

Engineering Funda

Structure of Solenoid and Generation of Magnetic Fields in Solenoid



Derivation of Magnetic Fields in Solenoid



⇒ As per Ampere Circuit Law.

$$\Rightarrow \oint \vec{H} \cdot d\vec{l} = I \quad \left[\vec{H} = \frac{\vec{B}}{\mu} \right]$$

$$\Rightarrow \oint \vec{B} \cdot d\vec{l} = \mu I$$

$$\Rightarrow \underbrace{\int_a^b \vec{B} \cdot d\vec{l}}_0 + \underbrace{\int_b^c \vec{B} \cdot d\vec{l}}_{90^\circ} + \underbrace{\int_c^d \vec{B} \cdot d\vec{l}}_{180^\circ} + \underbrace{\int_d^a \vec{B} \cdot d\vec{l}}_{-90^\circ} = \mu I$$

$$\Rightarrow B(L) = \mu I$$

$$\Rightarrow B = \frac{\mu I}{L}$$

- For N numbers of turns.

$$\Rightarrow B = \frac{\mu N I}{L}$$

$$\Rightarrow \boxed{B = \mu n I}$$

$$\left[n = \frac{N}{L} \text{ is numbers of turns per unit length} \right]$$

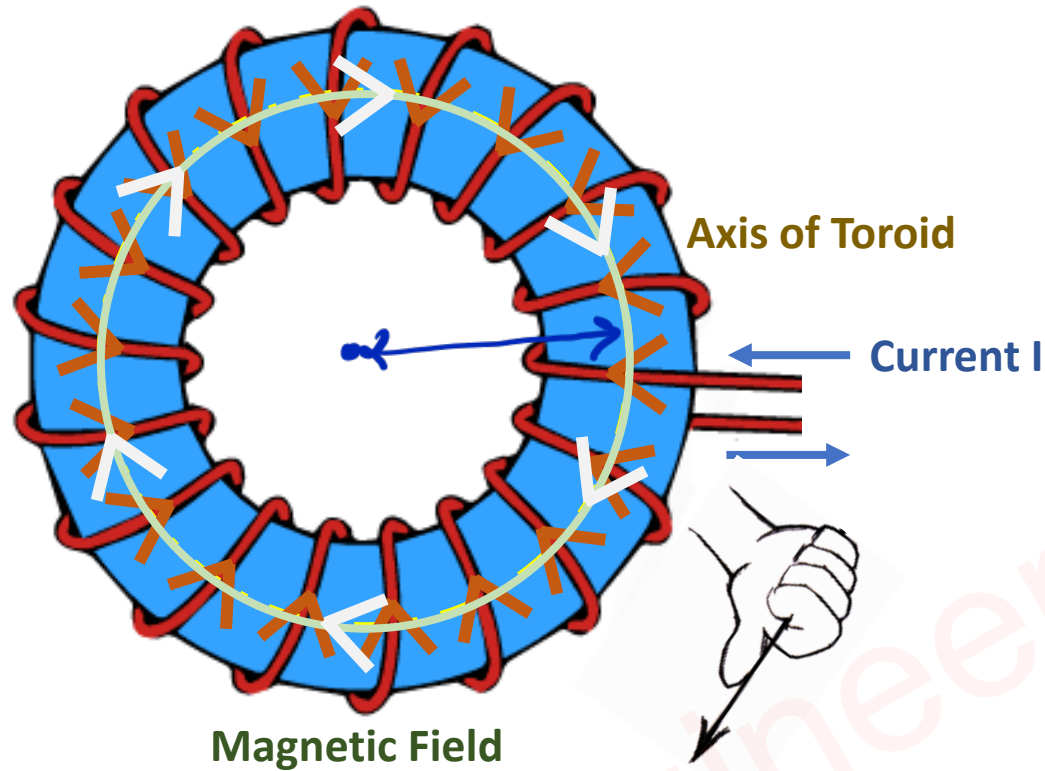
Toroid

Outlines

- Structure of Toroid
- Generation of Magnetic Field in Toroid
- Direction of Magnetic Field in Toroid
- Derivation of Magnetic Field in Toroid
- Solved Example of Toroid

Engineering Funda

Structure of Toroid and Magnetic Fields in Toroid



⇒ As per Ampere Circuit Law.

$$\Rightarrow \oint \vec{H} \cdot d\vec{l} = I \quad \left[\because \vec{H} = \frac{\vec{B}}{\mu} \right]$$

$$\Rightarrow \oint \vec{B} \cdot d\vec{l} = \mu I$$

$$\oint \vec{B} \cdot d\vec{l} = \mu I$$

$$\Rightarrow B(2\pi r) = \mu I$$

$$\Rightarrow B = \frac{\mu I}{2\pi r}$$

⇒ For N numbers of turns

$$\Rightarrow B = \frac{\mu N I}{2\pi r}$$

$$\left[n = \frac{N}{2\pi r}, \text{ Number of turns per unit length.} \right]$$

$$\Rightarrow \boxed{B = \mu n I}$$

Question – For Toroid, the radius is 0.2m, and the number of turns is 50. Find the magnetic field if we give 1A current to the Toroid.

$$\rightarrow \mu = \mu_0 \mu_r = 4\pi \times 10^{-7} \text{ SI}$$

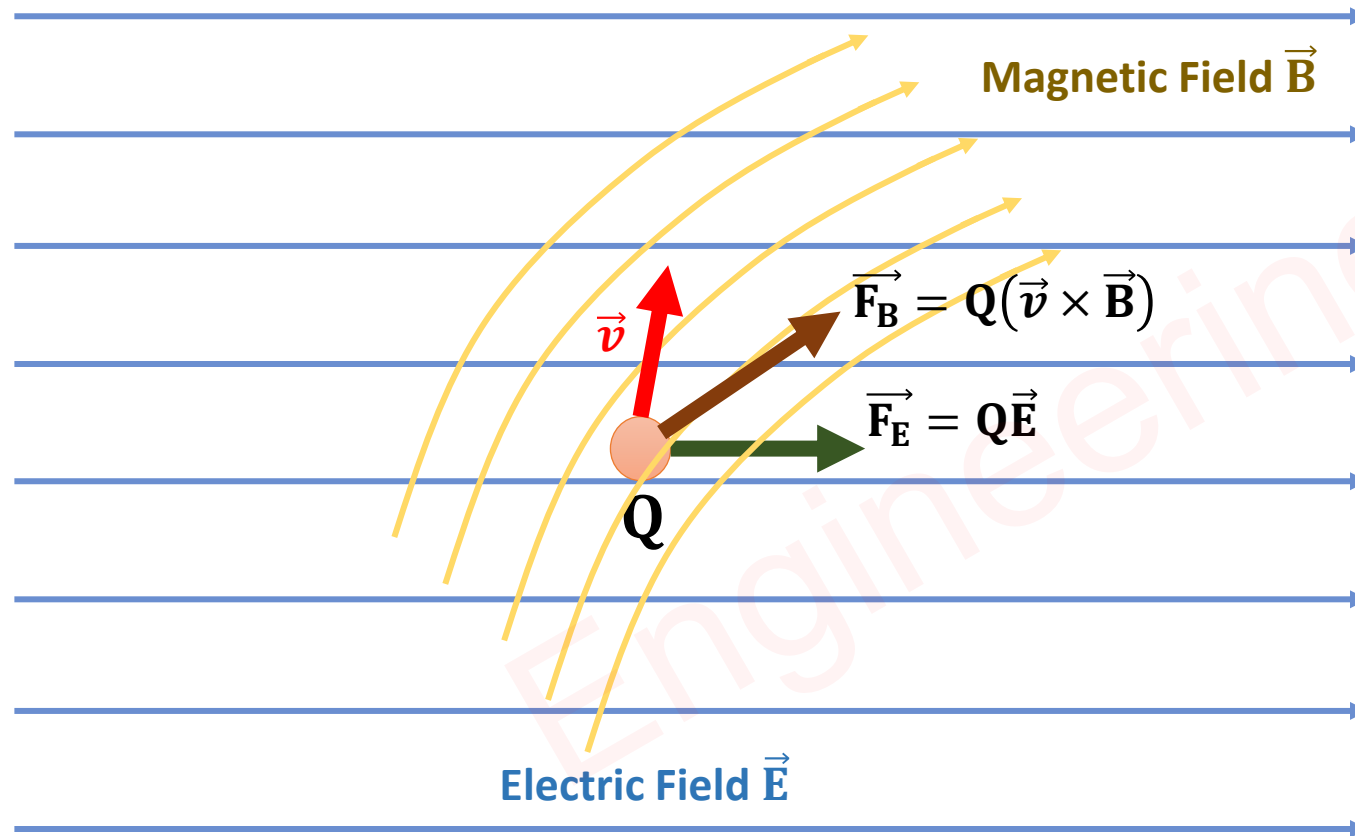
$$n = \frac{N}{2\pi r} = \frac{50}{2\pi \times 0.2} = 39.8$$

$$I = 1 \text{ A}$$

$$\begin{aligned} \rightarrow B &= \mu n I \\ &= 4\pi \times 10^{-7} \times 39.8 \times 1 \\ &= 5 \times 10^{-5} \text{ T} \\ &= 50 \text{ } \mu\text{T} \end{aligned}$$

Lorentz Force Equation

- Lorentz has explained the resultant force acting on moving charge due to Electric Field and Magnetic Field.



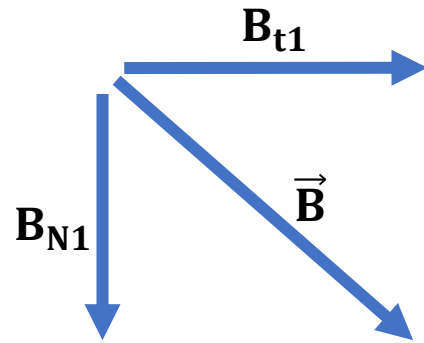
- Total Force on Moving charge is given by:

$$\Rightarrow \vec{F} = \vec{F}_E + \vec{F}_B = Q(\vec{E} + \vec{v} \times \vec{B})$$

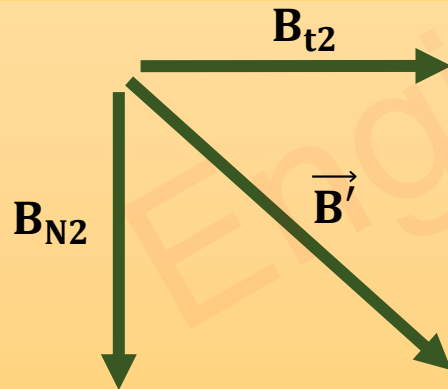
Magnetic Boundary Conditions

Basics of Magnetic Boundary Conditions

- Magnetic Boundary Conditions explain that Magnetic Field discontinuities occur when a Magnetic Field enters from one medium to another medium.

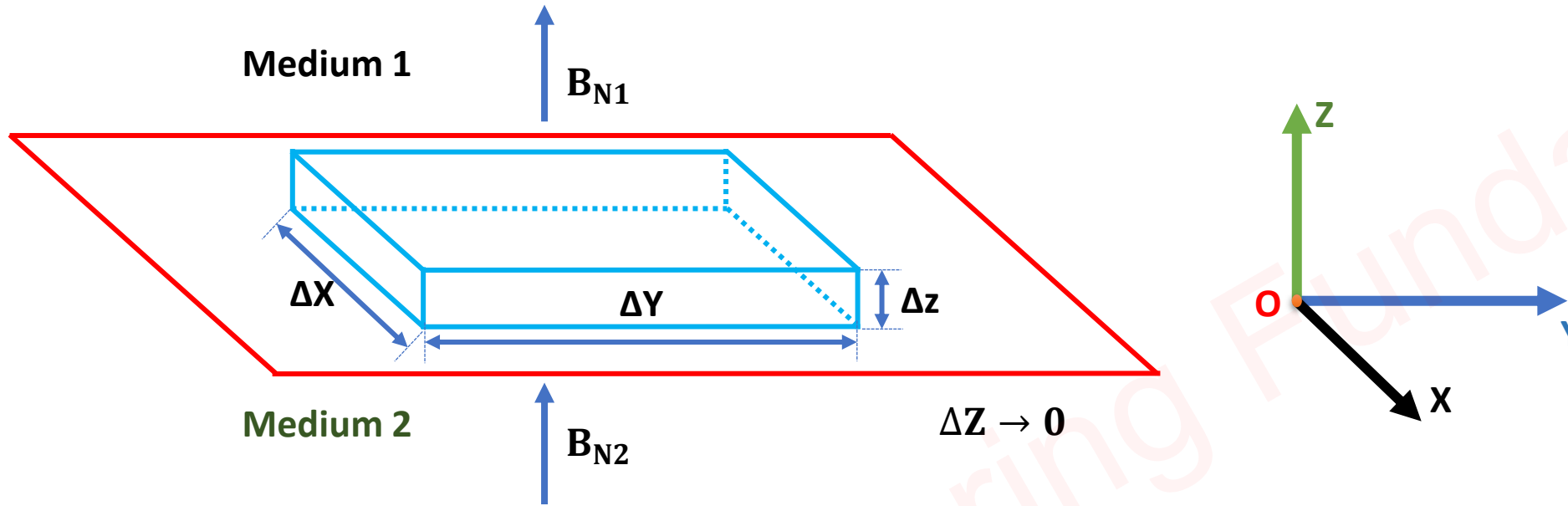


Medium - 1



Medium - 2

Magnetic Field Boundary Conditions for Normal Components



- As per Gauss Law of Magnetic Field:

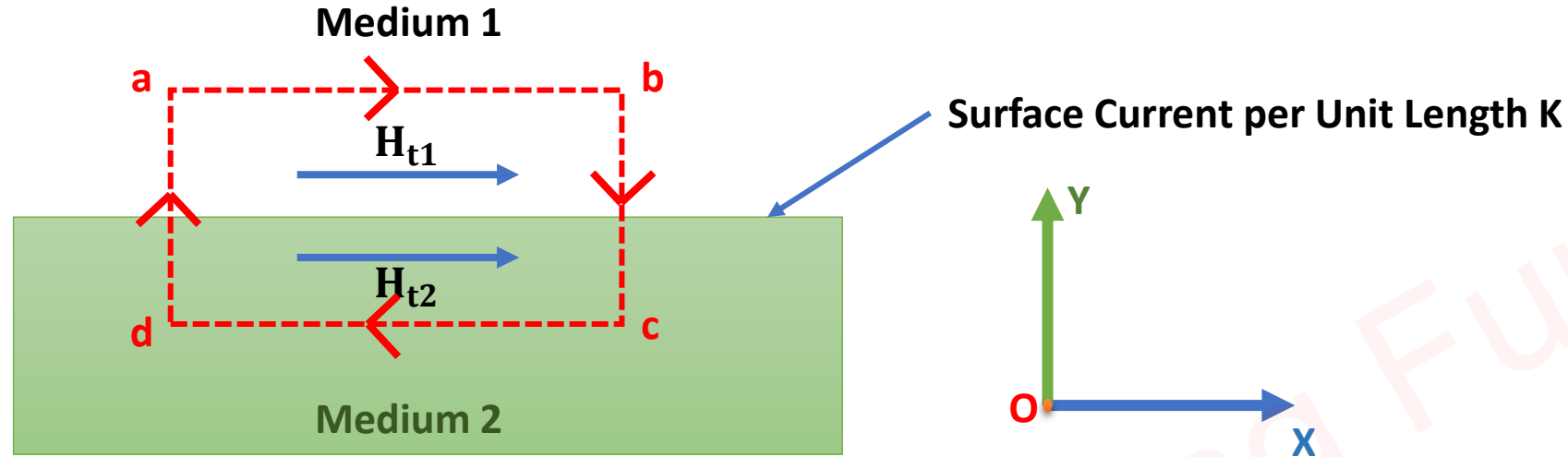
$$\Rightarrow \oint \vec{B} \cdot d\vec{S} = 0$$

$$\Rightarrow \int_{\text{Top}} \vec{B} \cdot d\vec{S} + \int_{\text{Bottom}} \vec{B} \cdot d\vec{S} + \int_{\text{Left}} \vec{B} \cdot d\vec{S} + \int_{\text{Right}} \vec{B} \cdot d\vec{S} + \int_{\text{Front}} \vec{B} \cdot d\vec{S} + \int_{\text{Back}} \vec{B} \cdot d\vec{S} = 0$$

$$\Rightarrow B_{n1} \Delta x \Delta y - B_{n2} \Delta x \Delta y = 0$$

$$\Rightarrow B_{n1} = B_{n2}$$

Magnetic Field Boundary Conditions for Tangential Components



- As per Ampere Circuit Law:

$$\Rightarrow \oint \vec{H} \cdot d\vec{l} = I = k\Delta x$$

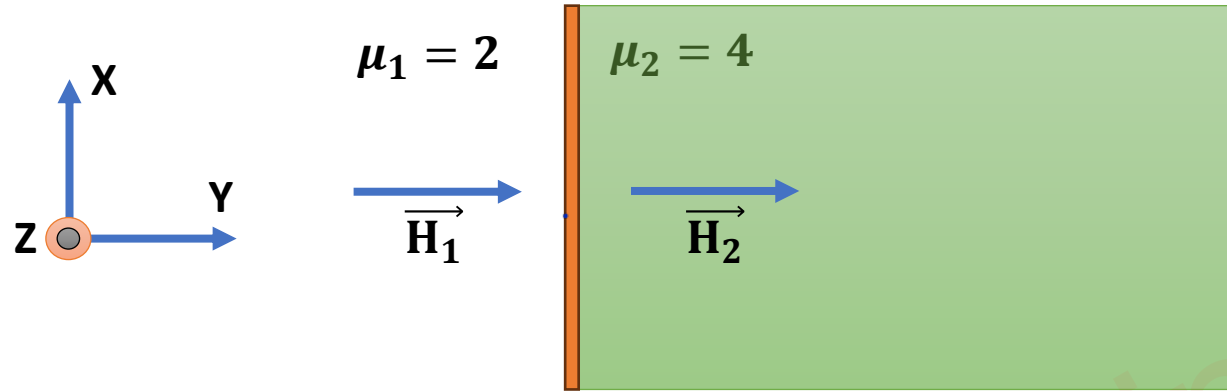
$$\Rightarrow \int_a^b \vec{H} \cdot d\vec{l} + \int_b^c \vec{H} \cdot d\vec{l} + \int_c^d \vec{H} \cdot d\vec{l} + \int_d^a \vec{H} \cdot d\vec{l} = k\Delta x$$

$$\Rightarrow H_{t1}\Delta x - H_{t2}\Delta x = k\Delta x$$

$$\Rightarrow H_{t1} - H_{t2} = k$$

Example of Magnetic Boundary Conditions

Question – $\vec{H}_1 = 4\hat{i}_x + 6\hat{i}_y + 2\hat{i}_z$, Then $\vec{H}_2 = ?$



→ Normal Component [y]

→ Tangential Components [x & z]

$$\vec{k} = 2\hat{i}_x + 4\hat{i}_z$$

→ Normal Component [y]

$$\Rightarrow B_{N1} = B_{N2}$$

$$\Rightarrow \cancel{\mu_0} \mu_1 H_{N1} = \cancel{\mu_0} \mu_2 H_{N2}$$

$$\Rightarrow 2 \times 6 = 4 H_{N2}$$

$$\Rightarrow H_{N2} = 3\hat{i}_y \text{ A/m}$$

→ Tangential Component [z]

$$\Rightarrow H_{T2} - H_{T1} = K$$

$$\Rightarrow H_{T2} = H_{T1} + K$$

$$= 2 + \frac{2\hat{i}_x}{\cancel{\mu_0}} \times \frac{\hat{i}_z}{\cancel{\mu_0}}$$

$\perp$ to H_{T1} $\uparrow$ Direction of propagation

$$= 2\hat{i}_z + 2\hat{i}_z = 4\hat{i}_z \text{ (A/m)}$$

→ Tangential Component [x]

$$\Rightarrow H_{T2} - H_{T1} = K$$

$$\overrightarrow{xy_2} \times \overrightarrow{xy_2}$$

$$\Rightarrow H_{T2} = H_{T1} + K$$

$$= 4 + \underline{4\hat{i}_2} \times \underline{\hat{i}_y}$$

↓
perp to H_{T1}

↑
Direction of propagation

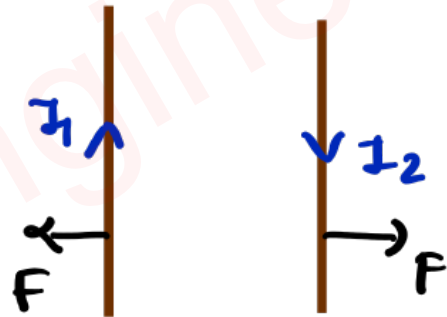
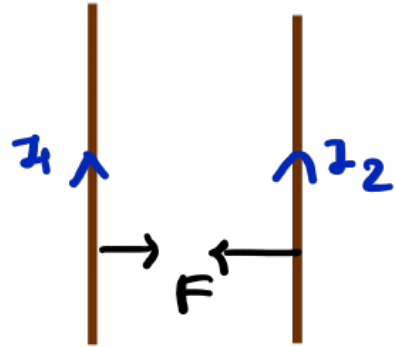
$$= 4\hat{i}_x - 4\hat{i}_x = 0$$

$$\rightarrow \vec{H}_2 = 0\hat{i}_x + 3\hat{i}_y + 4\hat{i}_z = 3\hat{i}_y + 4\hat{i}_z \text{ A/m}$$

Examples of Force by Magnetic Field

Question 1 – When a transformer winding suffers a short circuit, the adjoining turns of the same winding experience?

- ☒ a. An attractive force
- b. A repulsive force
- c. No force
- d. None of the above



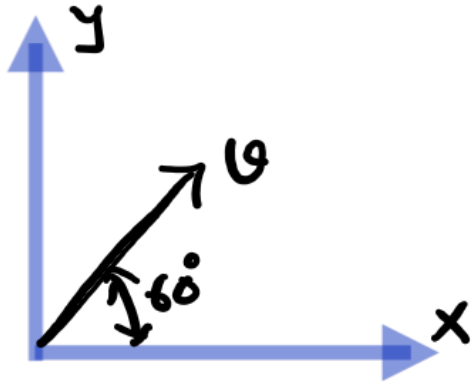
Question 2 – An electron moves in the XY plane with a speed of 10^6 m/sec. Its velocity vector makes an angle of 60° with x axis. A magnetic field of magnitude 10^{-2} T exists along the y axis. Compute the magnetic force exerted on the electron and its direction.

$$\rightarrow \vec{F}_m = q(\vec{v} \times \vec{B})$$

$$\rightarrow q = -1.6 \times 10^{-19} \text{ C}, \quad \vec{v} = v \cos 60^\circ \hat{i}_x + v \sin 60^\circ \hat{i}_y, \quad \vec{B} = 10^{-2} \hat{i}_y \text{ T}$$

$$= 10^6 \left(\frac{1}{2} \right) \hat{i}_x + 10^6 \left(\frac{\sqrt{3}}{2} \right) \hat{i}_y$$

$$= 0.5 \times 10^6 \hat{i}_x + 0.866 \times 10^6 \hat{i}_y$$



$$\rightarrow \vec{F}_m = q(\vec{v} \times \vec{B})$$

$$= -1.6 \times 10^{-19} (0.5 \times 10^6 \hat{i}_x + 0.866 \times 10^6 \hat{i}_y) \times 10^{-2} \hat{i}_y$$

$$= -1.6 \times 10^{-19} (0.5 \times 10^6 \times 10^{-2} \hat{i}_z)$$

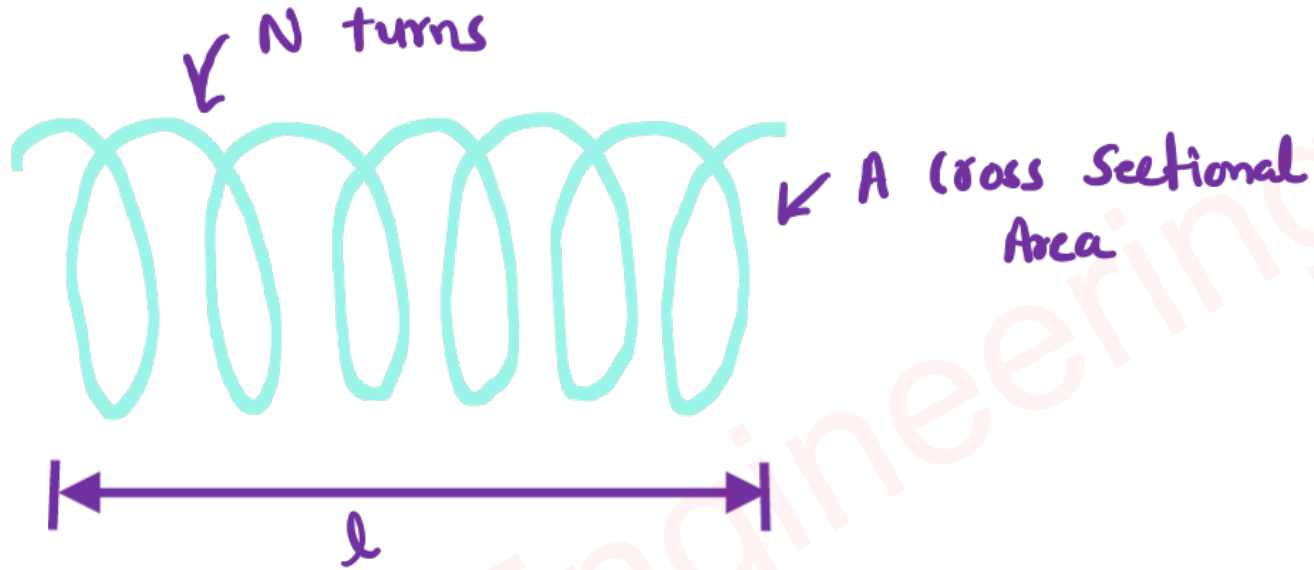
$$= -0.8 \times 10^{-15} \hat{i}_z \text{ N}$$

Question 3 – An electron with velocity v is placed in electric field E and magnetic field B . The force experienced by the electron is given by

$$\begin{aligned}\rightarrow \vec{F} &= \vec{F}_E + \vec{F}_B \\ &= q\vec{E} + q(\vec{v} \times \vec{B}) \\ &= q(\vec{E} + \vec{v} \times \vec{B})\end{aligned}$$

Examples of Inductance

Question 1 – The Inductance of a coil is proportional to the^{N²} of the number of turns, all over parameters remaining the same.



$$\rightarrow L = \frac{\mu_0 \mu_r N^2 A}{l} \Rightarrow L \propto N^2$$

Question 2 – A coil of 300 turns is wound on a non-magnetic core having a mean circumference of 300 mm and a cross-sectional area of 300 mm^2 . The inductance of the coil corresponding to a magnetizing current of 3A will be

- a. $37.68 \mu\text{H}$
- ☒ b. $113.04 \mu\text{H}$
- c. 37.68 nH
- d. 113.04 nH

$$\rightarrow N = 300$$

$$\mu_r = 1$$

$$l = 0.3 \text{ m}$$

$$A = 300 \times 10^{-6} \text{ m}^2$$

$$\rightarrow L = \frac{\mu_0 \mu_r N^2 A}{l} = \frac{4\pi \times 10^{-7} \times 1 \times 300^2 \times 300 \times 10^{-6}}{0.3}$$

$$= 113.04 \times 10^{-6} \text{ H}$$

$$= 113.04 \mu\text{H}$$

Question 3 – The inductance of a long solenoid of length 1000 mm wound uniformly with 3000 turns on a cylindrical paper tube of 60 mm diameter is

a. 3.2 μ H

b. 3.2 mH

✓ c. 32 mH

d. 3.2 H

$$\rightarrow l = 1 \text{ m}$$

$$N = 3000$$

$$A = \pi r^2 = \pi \times 30^2 \times 10^{-6} \text{ m}^2$$

$$\rightarrow L = \frac{\mu_0 \mu_r N^2 A}{l}$$

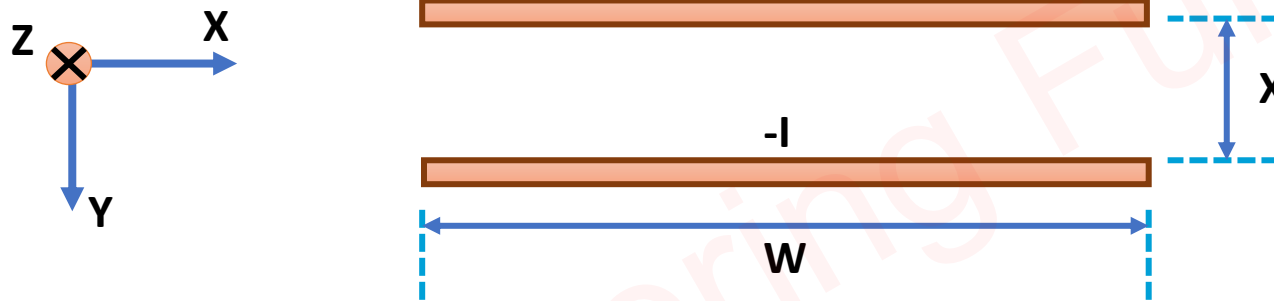
$$= \frac{4\pi \times 10^{-7} \times 1 \times 3000^2 \times \pi \times 30^2 \times 10^{-6}}{1}$$

$$= 32 \times 10^{-3} \text{ H}$$

$$= 32 \text{ mH}$$

Question 4 – Two infinite strips of width w in x direction as shown in figure are carrying forward and return current $+I$ and $-I$ in the z direction. The strips are separated by a distance x . The inductance per unit length of the configuration is measured to be L (H/m). If the distance of separation between the strips is now reduce to $(x/2)$. The inductance per unit length of the configuration is $L/2$ H/m

- a. $2L$
- b. $L/4$
- c. $L/2$
- d. $4L$



→ $L \propto$ Spacing bet.ⁿ two parallel strips.

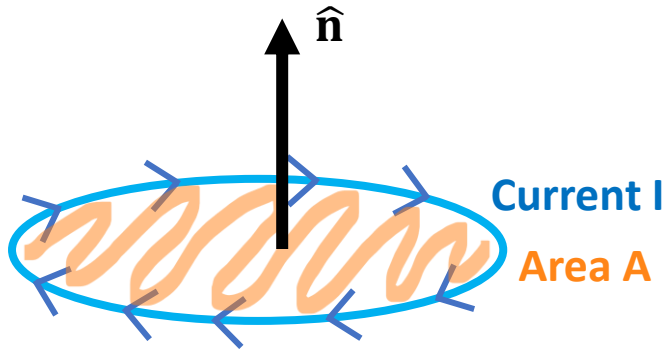
Magnetic Properties of Material

Outlines

- Magnetic Dipole Moment
- Magnetization
- Magnetic Susceptibility
- Relative Permeability

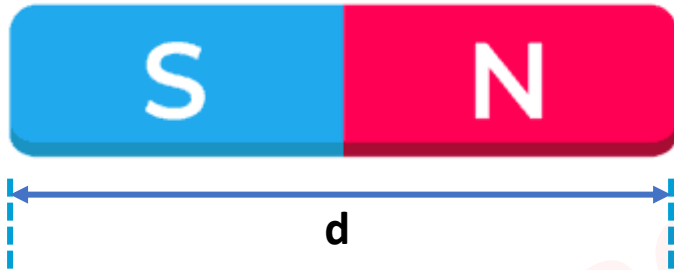
Engineering Funda

Magnetic Dipole Moment



- Magnetic Dipole Moment $\vec{m}$ is given by:

$$\Rightarrow \vec{m} = IA \hat{n} [\text{Am}^2]$$



- Magnetic Dipole Moment m is given by:

$$\Rightarrow m = pd$$

- Where, p is pole strength.
- d is distance between two poles.

Magnetization M

- Magnetization M depends on material.
- Magnetization M is Magnetic Dipole Moment per unit Volume.

$$\Rightarrow M = \frac{m}{V} \text{ [A/m]}$$

- The relation between Magnetization $\vec{M}$ and Magnetic Susceptibility χ_m is given by:

$$\Rightarrow \vec{M} = \chi_m \vec{H}$$

- The Total Magnetic Field $\vec{B}$ due to Magnetization $\vec{M}$ and Magnetic field Intensity $\vec{H}$ is given by:

$$\Rightarrow \vec{B} = \mu_0 (\vec{H} + \vec{M})$$

$$\Rightarrow \vec{B} = \mu_0 (\vec{H} + \chi_m \vec{H}) = \mu_0 (1 + \chi_m) \vec{H} = \mu_0 \mu_r \vec{H}$$

- Hence, Relative Permeability μ_r and Magnetic Susceptibility χ_m are related as per:

$$\Rightarrow \mu_r = 1 + \chi_m$$